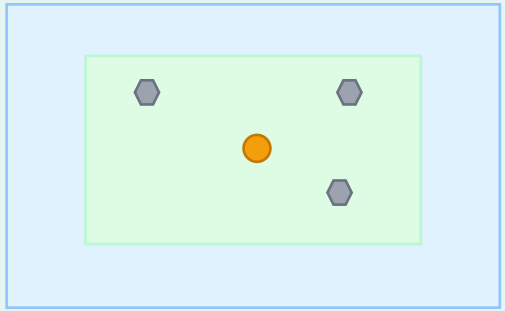
De escena a juego interactivo: scripting e Input System

De escena a juego interactivo: scripting e Input System

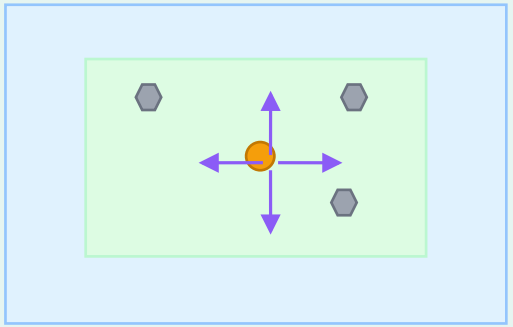
La presentación anterior nos dejó:

`Arena01`
`Fondo` · `Jugador` · `3 Rocas` · `Camera` · `Roca.prefab`



Ahora añadimos

- **script del Jugador**
- **entrada del usuario**
- **movimiento visible**



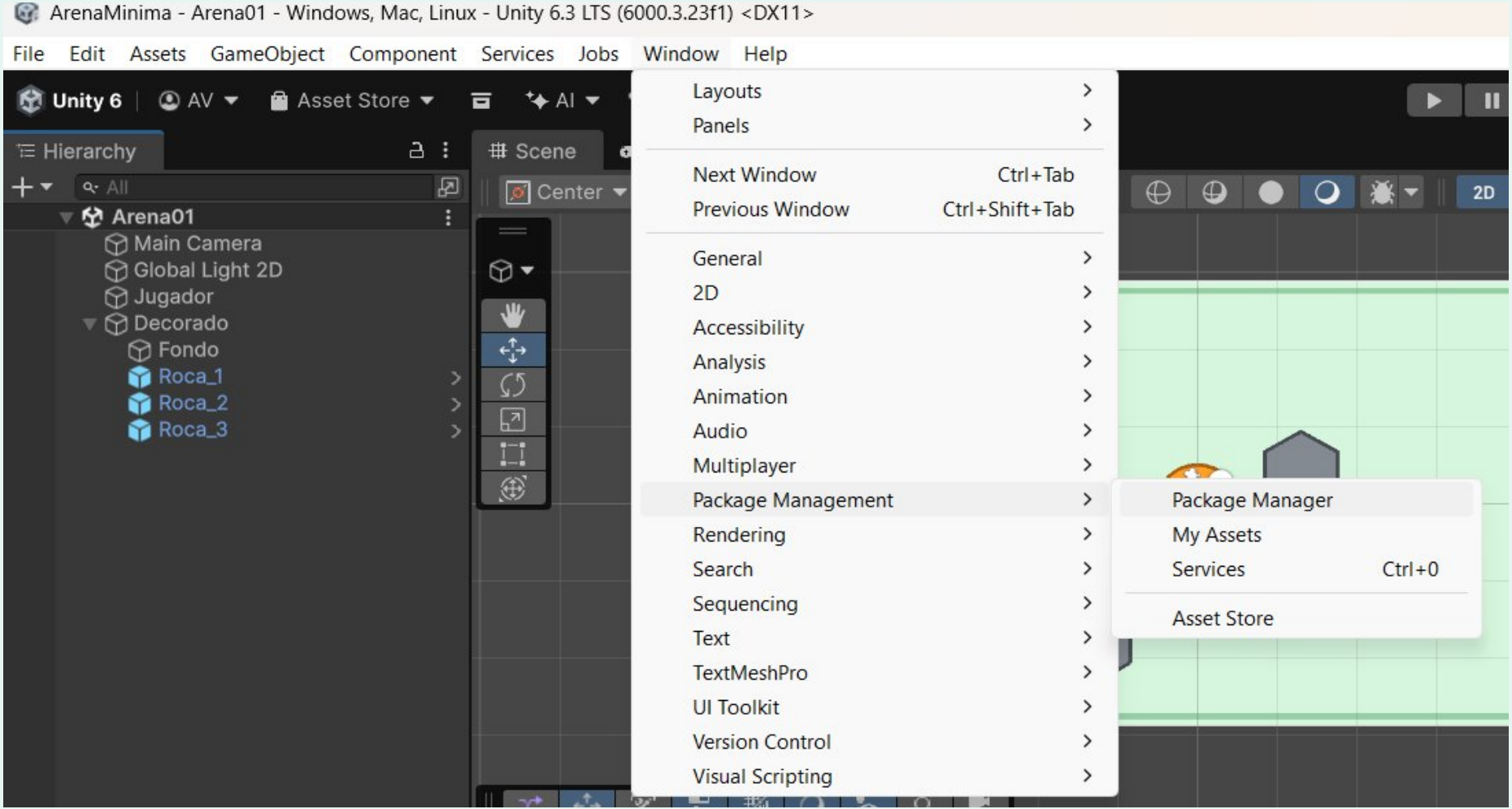
Hoy:

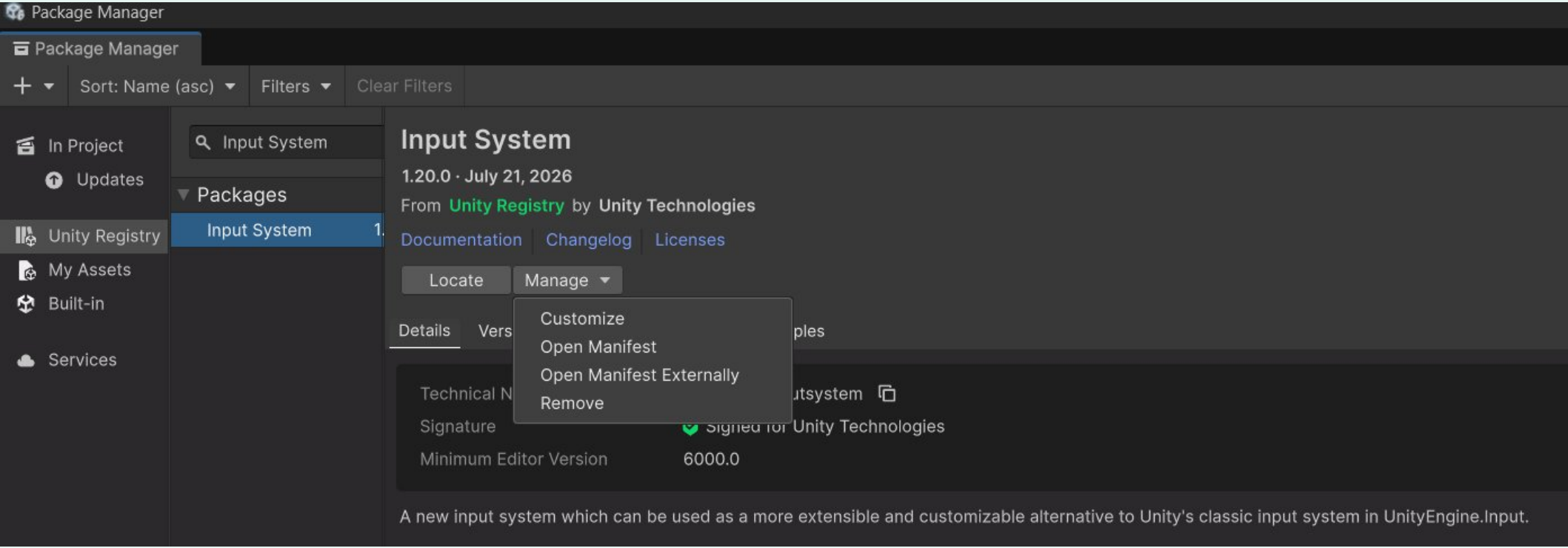
scripting + Input System
movimiento programado
sin física 2D

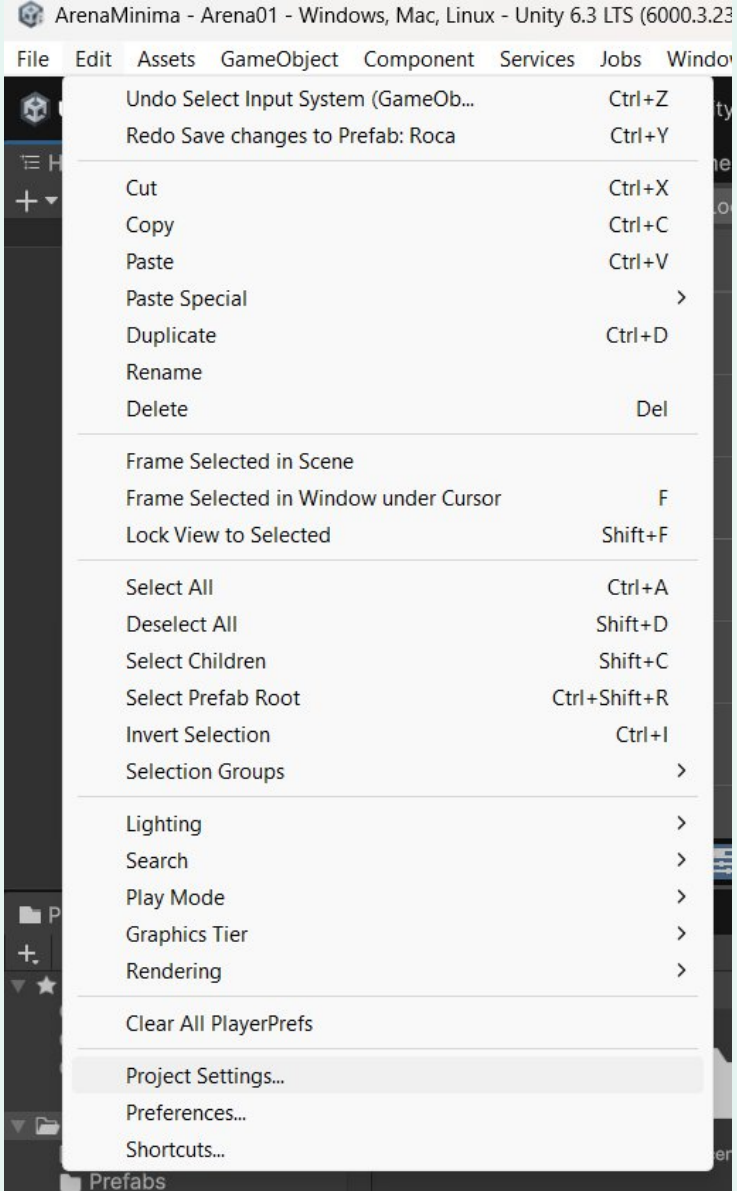
sin Rigidbody2D · sin Collider2D · sin colisiones

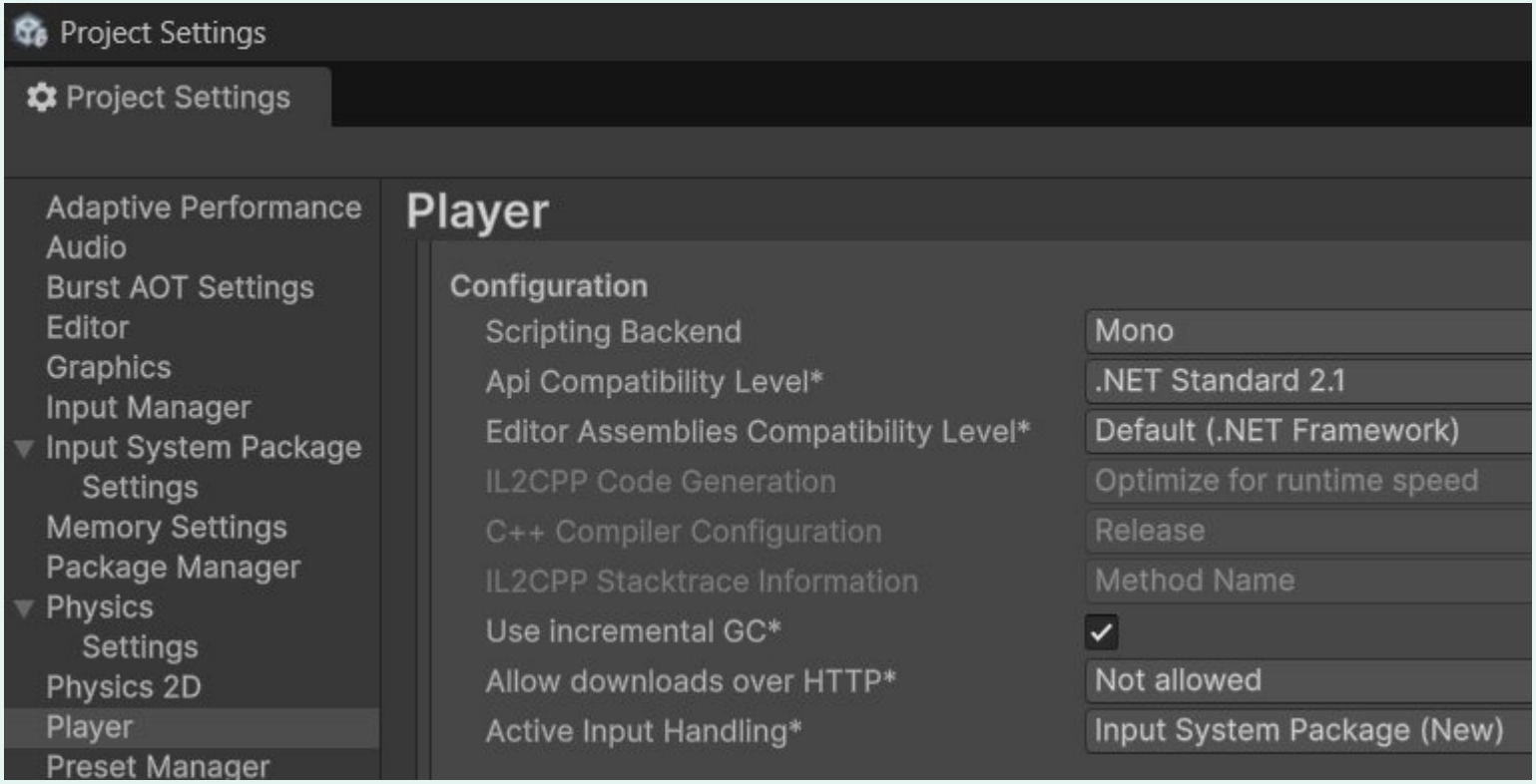
Input System

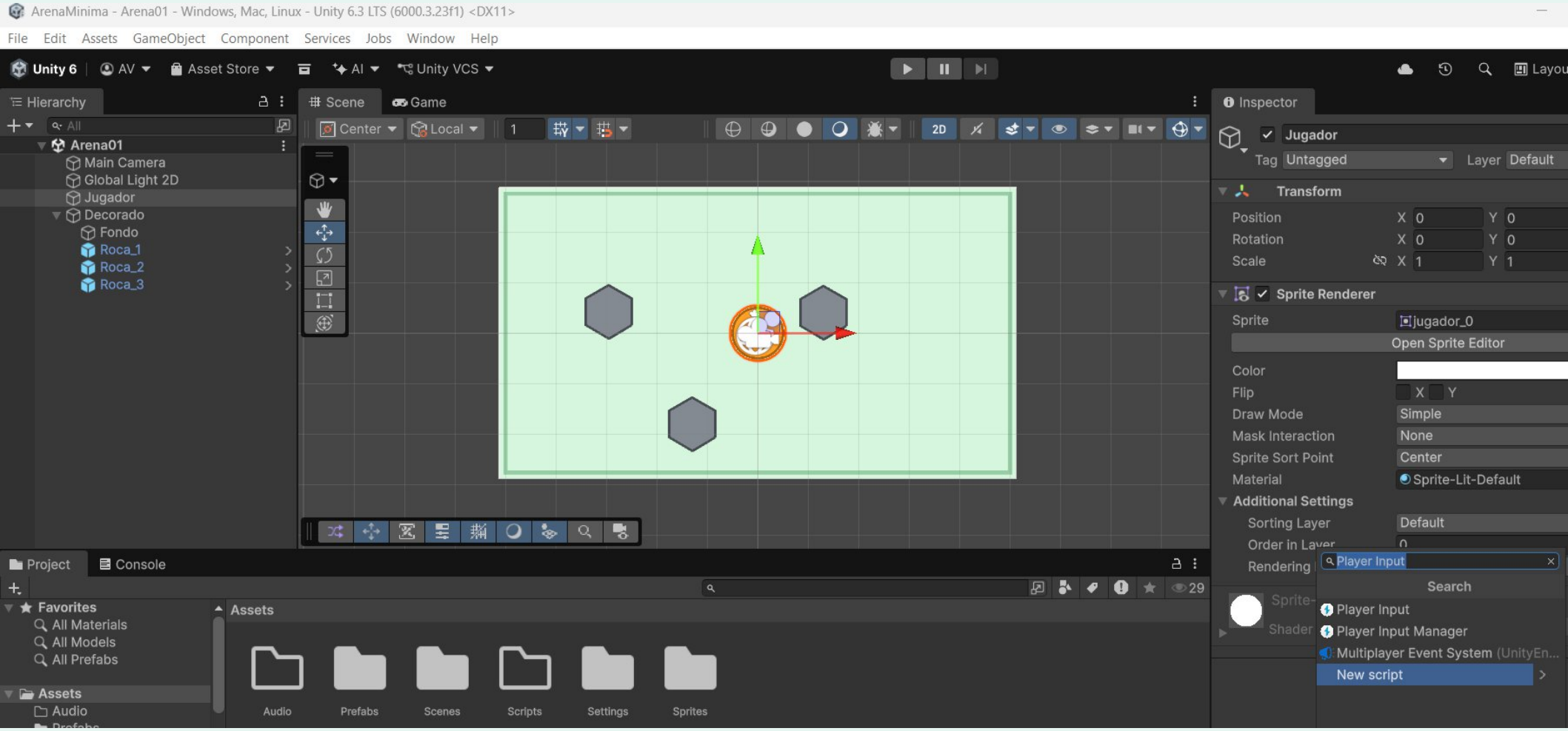
- Input System es el sistema moderno de Unity para recibir entrada del usuario (teclado, ratón, …).
- Instalamos Input System / Comprobar que está instalado.
- Activamos Input System / Comprobar que está instalado.
- Comprobamos que no aparecen errores en Console.
- Comprobamos que podemos añadir Components de tipo Player Input
- Comprobamos que podemos crear Assets de tipo Input Actions

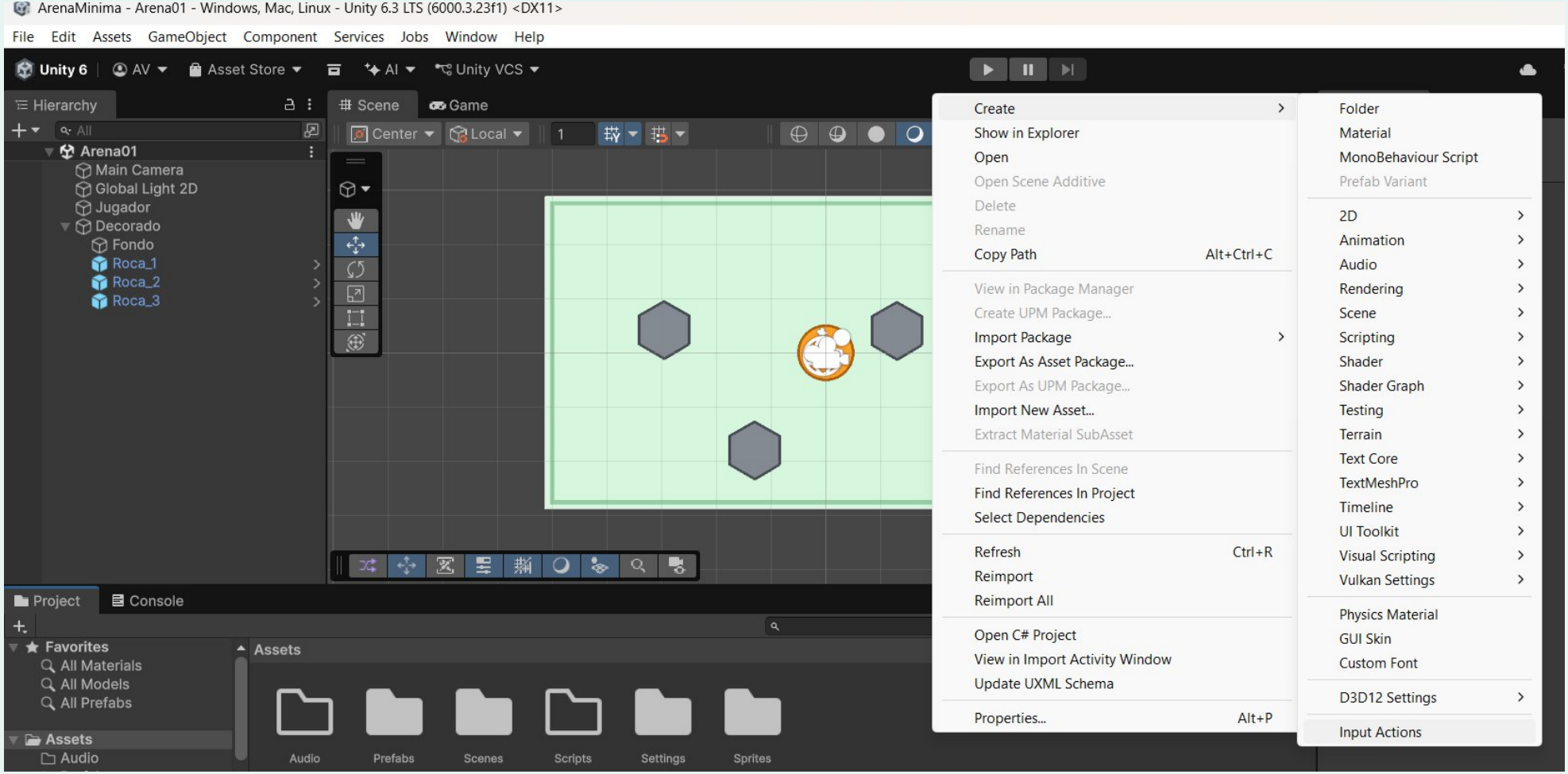

ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1) <DX11>
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Next Window Ctrl+Tab
Previous Window Ctrl+Shift+Tab
General
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Accessibility
Analysis
Animation
Audio
Multiplayer
Package Management
Rendering
Search
Sequencing
Text
TextMeshPro
UI Toolkit
Version Control
Visual Scripting
Package Manager
My Assets
Services Ctrl+0
Asset Store


Package Manager
Package Manager
Sort: Name (asc)
Filters
Clear Filters
In Project
Updates
Unity Registry
My Assets
Built-in
Services
Input System
Packages
Input System
Input System
1.20.0 · July 21, 2026
From Unity Registry by Unity Technologies
Documentation
Changelog
Licenses
Locate
Manage
Customize
Open Manifest
Open Manifest Externally
Remove
Details
Technical N
utsystem
Signature
Signed for Unity Technologies
Minimum Editor Version
6000.0
A new input system which can be used as a more extensible and customizable alternative to Unity's classic input system in UnityEngine.Input.


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23
File
Edit
Assets
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Services
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Undo Select Input System (GameOb... Ctrl+Z
Redo Save changes to Prefab: Roca Ctrl+Y
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Paste Special
Duplicate Ctrl+D
Rename
Delete Del
Frame Selected in Scene
Frame Selected in Window under Cursor F
Lock View to Selected Shift+F
Select All Ctrl+A
Deselect All Shift+D
Select Children Shift+C
Select Prefab Root Ctrl+Shift+R
Invert Selection Ctrl+I
Selection Groups
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Graphics Tier
Rendering
Clear All PlayerPrefs
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Preferences...
Shortcuts...
Prefabs


Project Settings
Project Settings
Adaptive Performance
Audio
Burst AOT Settings
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Graphics
Input Manager
Input System Package
Settings
Memory Settings
Package Manager
Physics
Settings
Physics 2D
Player
Preset Manager
Player
Configuration
Scripting Backend
Mono
Api Compatibility Level*
.NET Standard 2.1
Editor Assemblies Compatibility Level*
Default (.NET Framework)
IL2CPP Code Generation
Optimize for runtime speed
C++ Compiler Configuration
Release
IL2CPP Stacktrace Information
Method Name
Use incremental GC*
Allow downloads over HTTP*
Not allowed
Active Input Handling*
Input System Package (New)


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1) <DX11>
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Draw Mode
Simple
Mask Interaction
None
Sprite Sort Point
Center
Material
Sprite-Lit-Default
Additional Settings
Sorting Layer
Default
Order in Layer
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Search
Player Input
Player Input Manager
Multiplayer Event System
New script
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Sprites


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All Materials
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Assets
Audio
Assets
Audio Prefabs Scenes Scripts Settings Sprites
Create
Show in Explorer
Open
Open Scene Additive
Delete
Rename
Copy Path Alt+Ctrl+C
View in Package Manager
Create UPM Package...
Import Package
Export As Asset Package...
Export As UPM Package...
Import New Asset...
Extract Material SubAsset
Find References In Scene
Find References In Project
Select Dependencies
Refresh Ctrl+R
Reimport
Reimport All
Open C# Project
View in Import Activity Window
Update UXML Schema
Properties... Alt+P
Folder
Material
MonoBehaviour Script
Prefab Variant
2D
Animation
Audio
Rendering
Scene
Scripting
Shader
Shader Graph
Testing
Terrain
Text Core
TextMeshPro
Timeline
UI Toolkit
Visual Scripting
Vulkan Settings
Physics Material
GUI Skin
Custom Font
D3D12 Settings
Input Actions

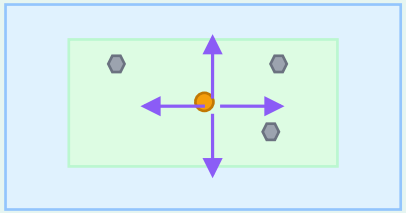
Antes de programar: definir qué queremos que ocurra

Comportamiento deseado → reglas programables

Inicio	Puede hacer	Acontecimientos futuros	Éxito futuro	Fracaso futuro	Final → reinicio
`Jugador` comienza en `Arena01`	moverse en 2D	podrán modificar el estado	alcanzar una meta	activar derrota	volver al estado inicial

P2 implementa solo: **entrada** → **dirección** → **desplazamiento**

Definir la mecánica ≠ implementarla completa hoy



`Arena01` como escenario de continuidad

Del archivo C# al Component del Jugador

Project

PlayerController.cs

asset de script
contiene la clase:

PlayerController :
MonoBehaviour

lo añadimos a `Jugador`

Inspector

`Jugador`

Transform
SpriteRenderer
PlayerController (Script)

Component del `Jugador`
instancia de esa clase

Hierarchy → GameObjects
Inspector → Components

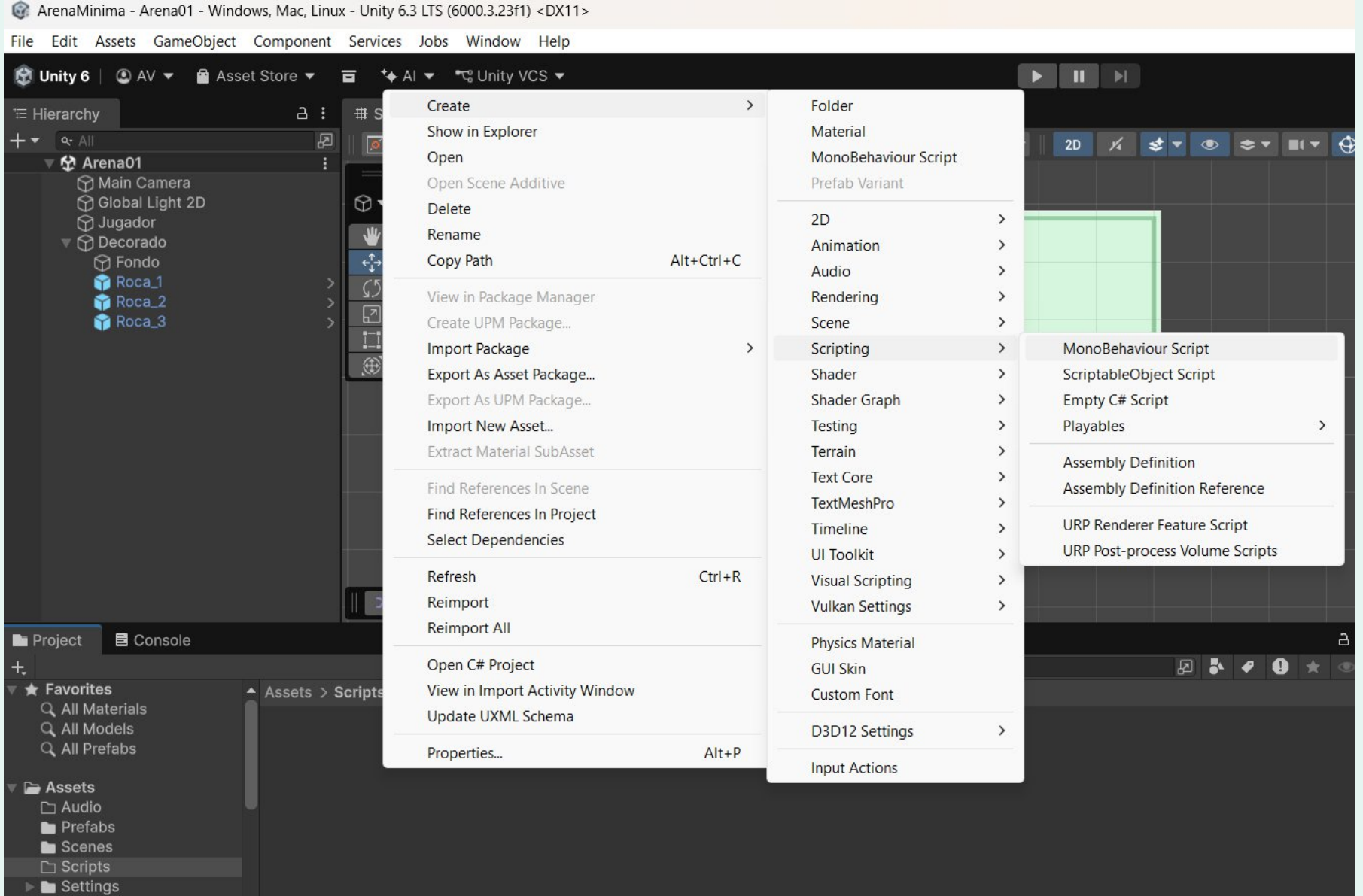
No son lo mismo

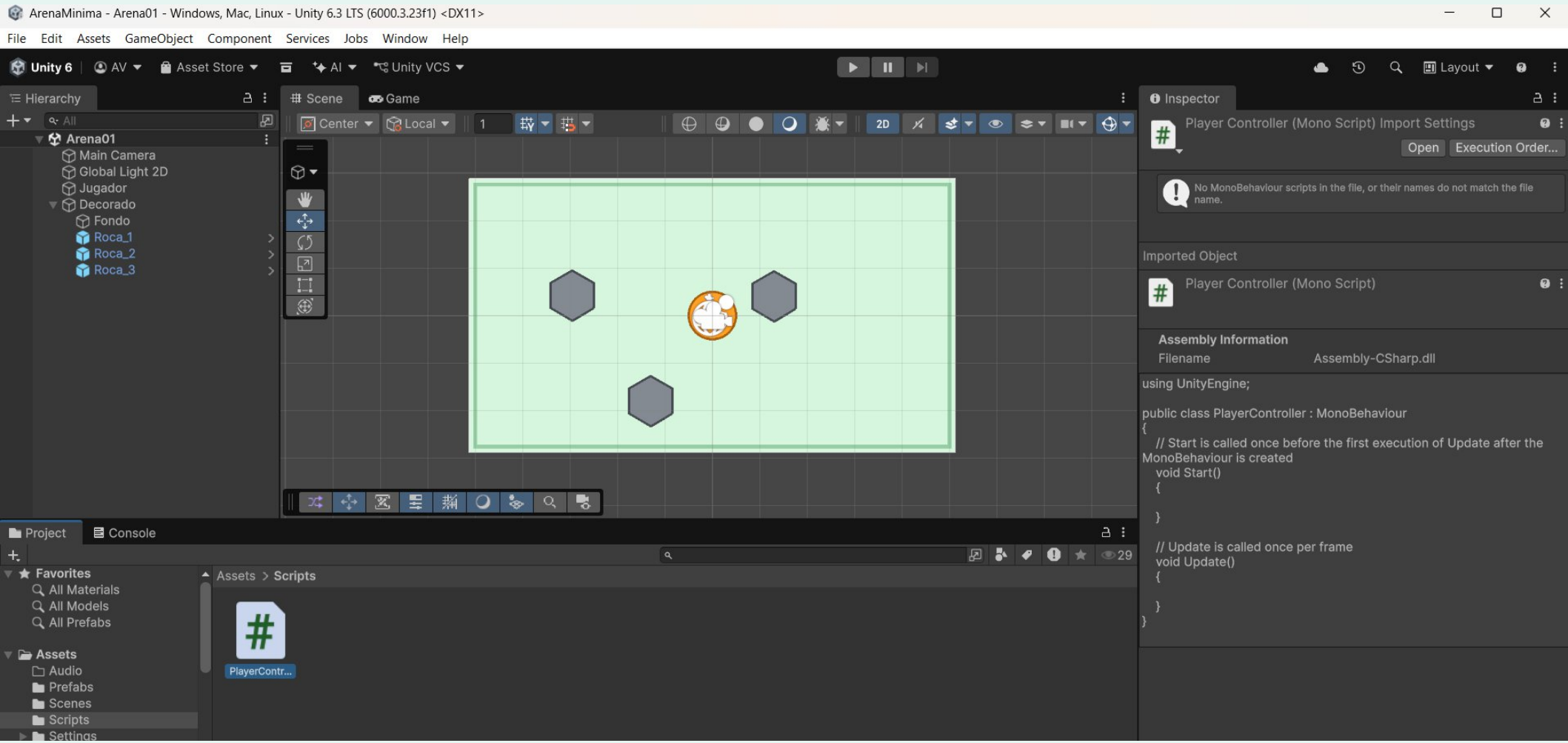
- archivo `.cs` ≠ GameObject
- script asset ≠ Component
- `MonoBehaviour` ≠ GameObject
- Component ≠ hijo en Hierarchy

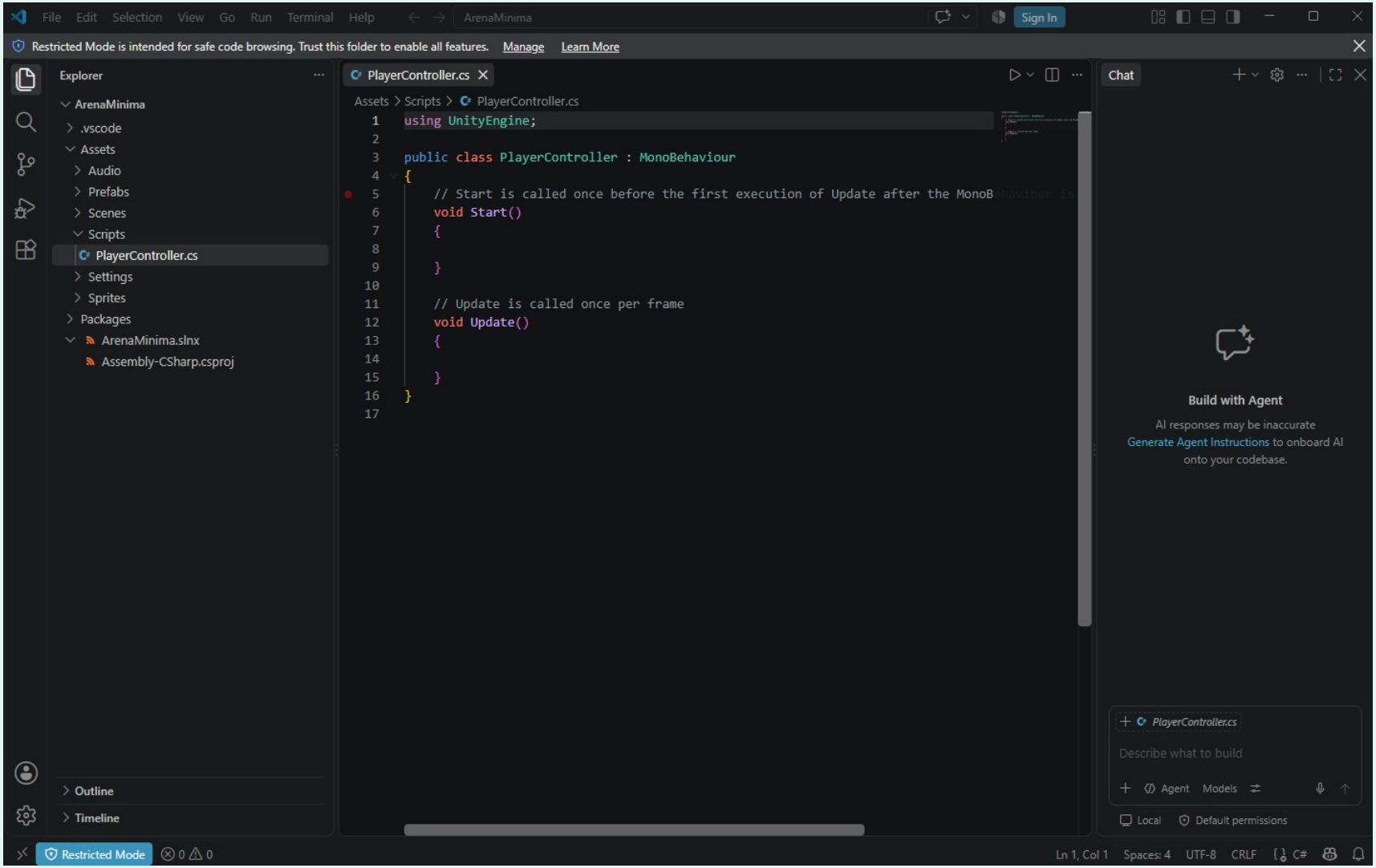
P1: funcionalidad por composición
P2: nuestro script se suma como Component

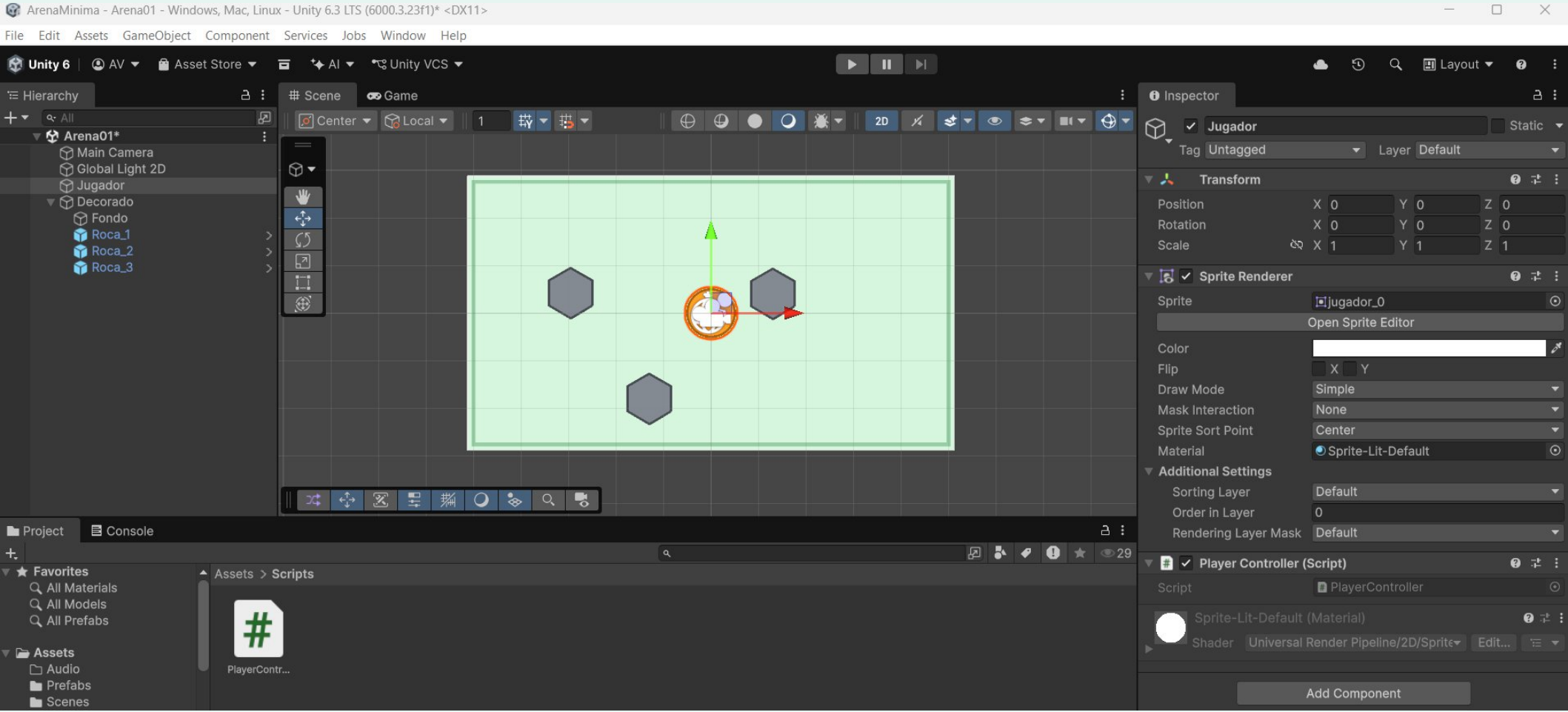
Del archivo C# al Component del Jugador

- Creamos el script PlayerController (MonoBehaviour Script) en Assets/Scripts.
- Lo abrimos con el editor de código.
- Añadimos el Script al GameObject Jugador y lo vemos en el Inspector.


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1) <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Unity 6 AV Asset Store AI Unity VCS
Hierarchy
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Main Camera
Global Light 2D
Jugador
Decorado
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Create
Show in Explorer
Open
Open Scene Additive
Delete
Rename
Copy Path Alt+Ctrl+C
View in Package Manager
Create UPM Package...
Import Package
Export As Asset Package...
Export As UPM Package...
Import New Asset...
Extract Material SubAsset
Find References In Scene
Find References In Project
Select Dependencies
Refresh Ctrl+R
Reimport
Reimport All
Open C# Project
View in Import Activity Window
Update UXML Schema
Properties... Alt+P
Folder
Material
MonoBehaviour Script
Prefab Variant
2D
Animation
Audio
Rendering
Scene
Scripting
Shader
Shader Graph
Testing
Terrain
Text Core
TextMeshPro
Timeline
UI Toolkit
Visual Scripting
Vulkan Settings
Physics Material
GUI Skin
Custom Font
D3D12 Settings
Input Actions
MonoBehaviour Script
ScriptableObject Script
Empty C# Script
Playables
Assembly Definition
Assembly Definition Reference
URP Renderer Feature Script
URP Post-process Volume Scripts
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All Prefabs
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Settings
Assets > Scripts


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Roca_1
Roca_2
Roca_3
Scene Game
Center Local 1 2D
Inspector
Player Controller (Mono Script) Import Settings
Open Execution Order...
No MonoBehaviour scripts in the file, or their names do not match the file name.
Imported Object
Player Controller (Mono Script)
Assembly Information
Filename Assembly-CSharp.dll
using UnityEngine;
public class PlayerController : MonoBehaviour
{
// Start is called once before the first execution of Update after the MonoBehaviour is created
void Start()
{
}
// Update is called once per frame
void Update()
{
}
}
Project Console
Favorites
All Materials
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All Prefabs
Assets
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Scenes
Scripts
Settings
Assets > Scripts
PlayerContr...

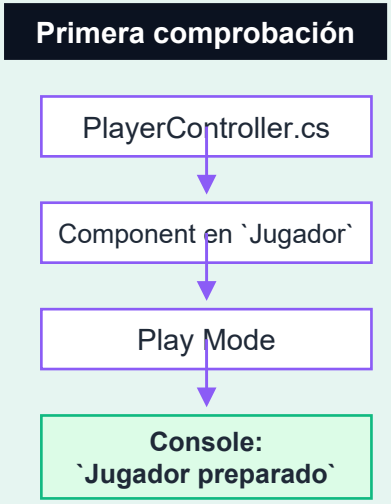

File Edit Selection View Go Run Terminal Help
ArenaMinima
Sign In
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
Explorer
ArenaMinima
.vscode
Assets
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PlayerController.cs
Settings
Sprites
Packages
ArenaMinima.slnx
Assembly-CSharp.csproj
Outline
Timeline
PlayerController.cs
Assets > Scripts > PlayerController.cs
1 using UnityEngine;
2
3 public class PlayerController : MonoBehaviour
4 {
5 // Start is called once before the first execution of Update after the MonoB
6 void Start()
7 {
8
9 }
10
11 // Update is called once per frame
12 void Update()
13 {
14
15 }
16 }
17
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
PlayerController.cs
Describe what to build
Agent
Models
Local
Default permissions
Restricted Mode
0 0
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
C#


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1)* <DX11>
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Jugador
Static
Tag Untagged Layer Default
Transform
Position X 0 Y 0 Z 0
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Sprite Renderer
Sprite jugador_0
Open Sprite Editor
Color
Flip X Y
Draw Mode Simple
Mask Interaction None
Sprite Sort Point Center
Material Sprite-Lit-Default
Additional Settings
Sorting Layer Default
Order in Layer 0
Rendering Layer Mask Default
Player Controller (Script)
Script PlayerController
Sprite-Lit-Default (Material)
Shader Universal Render Pipeline/2D/Sprite Edit...
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PlayerContr...

Primer script: comprobar que Unity lo ejecuta

```
using UnityEngine;

public class PlayerController :
MonoBehaviour
{
    private void Start()
    {
        Debug.Log("Jugador
preparado");
    }
}
```

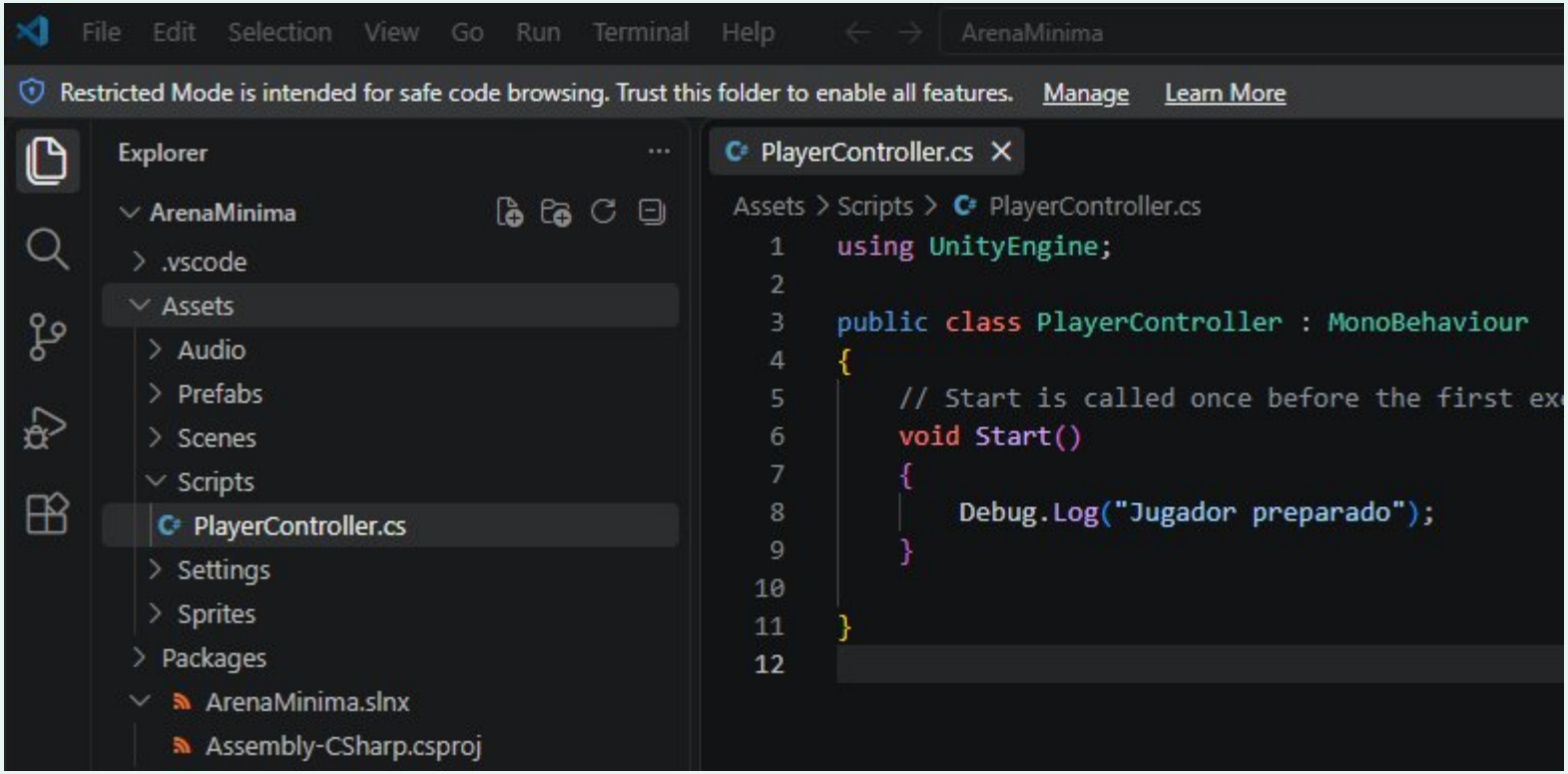


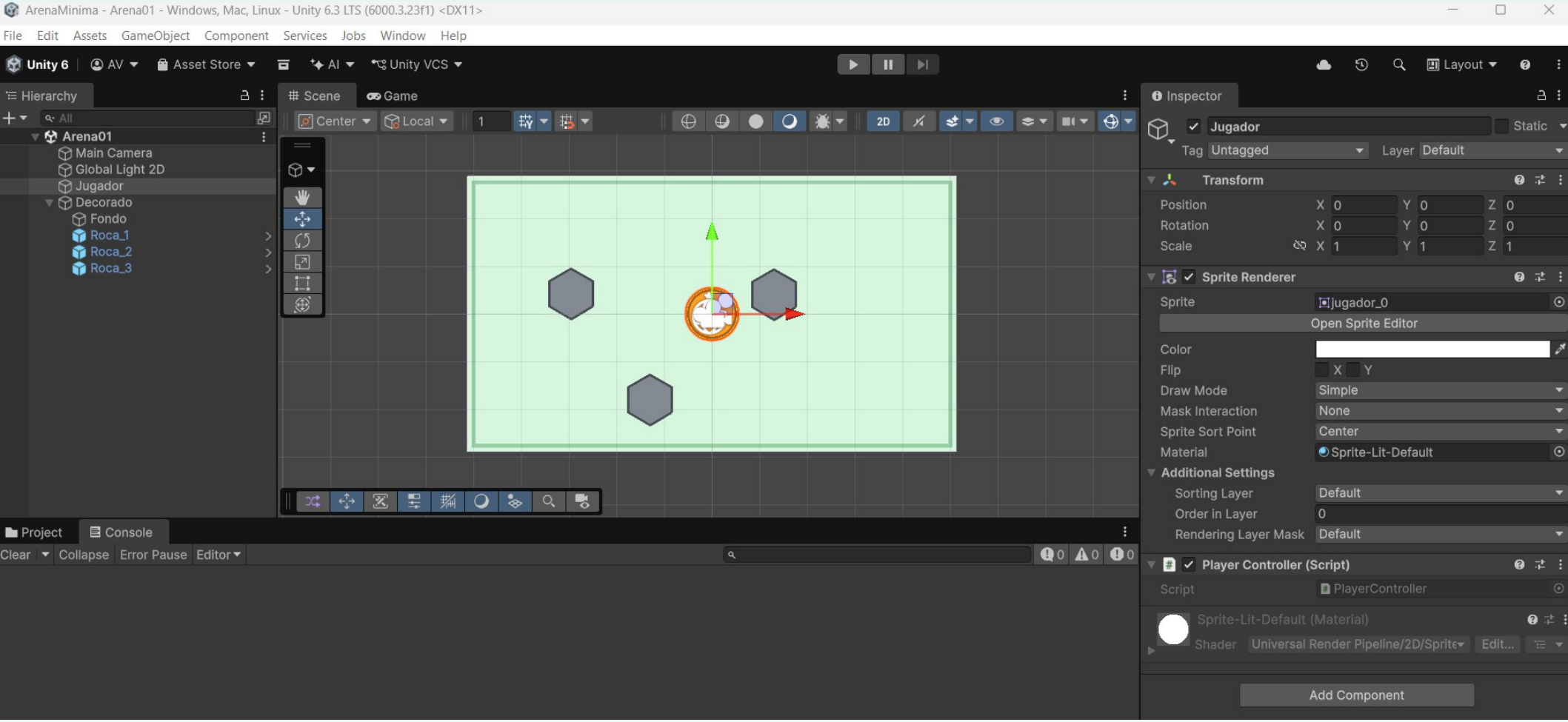
`Start` ≠ constructor de C#

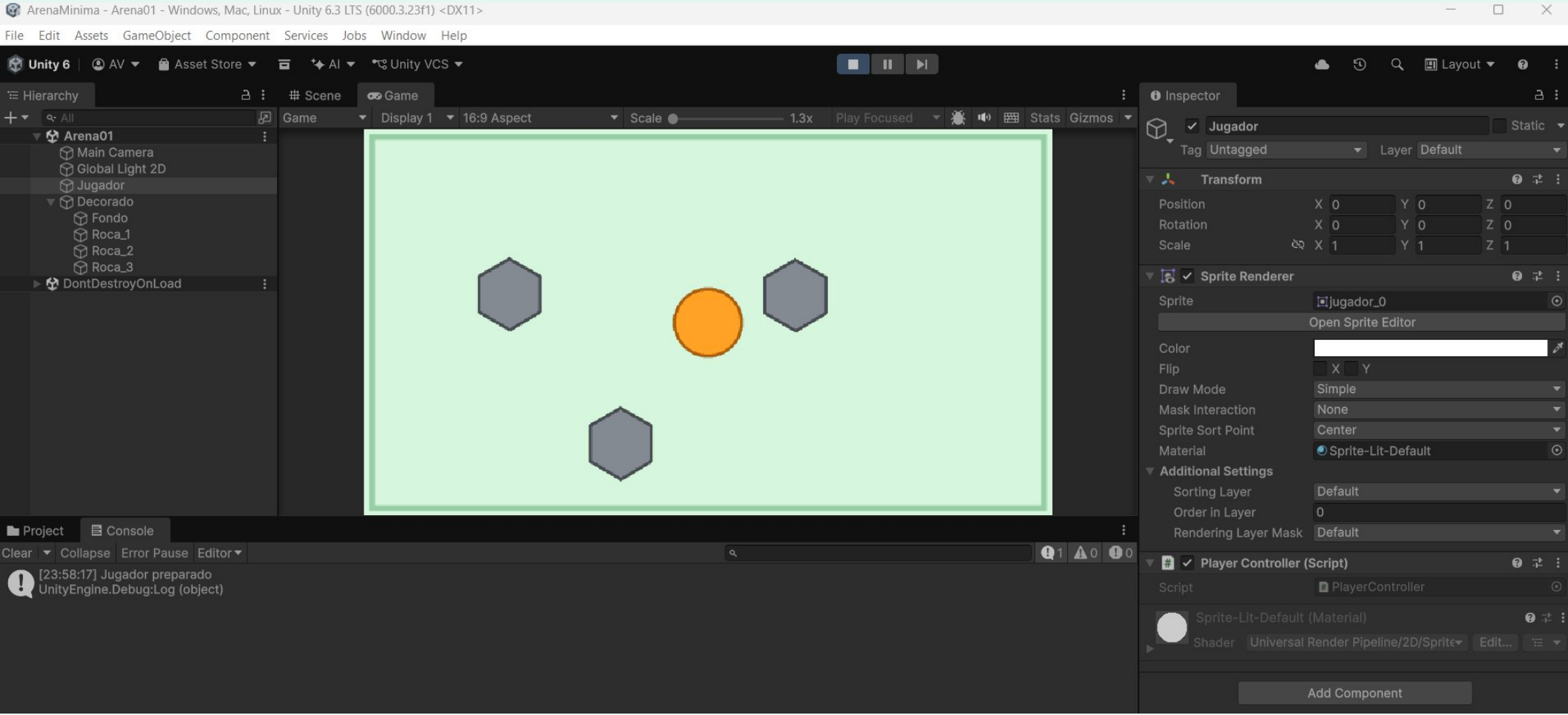
Unity lo llama una vez, antes del primer `Update` de este Component cuando está habilitado.

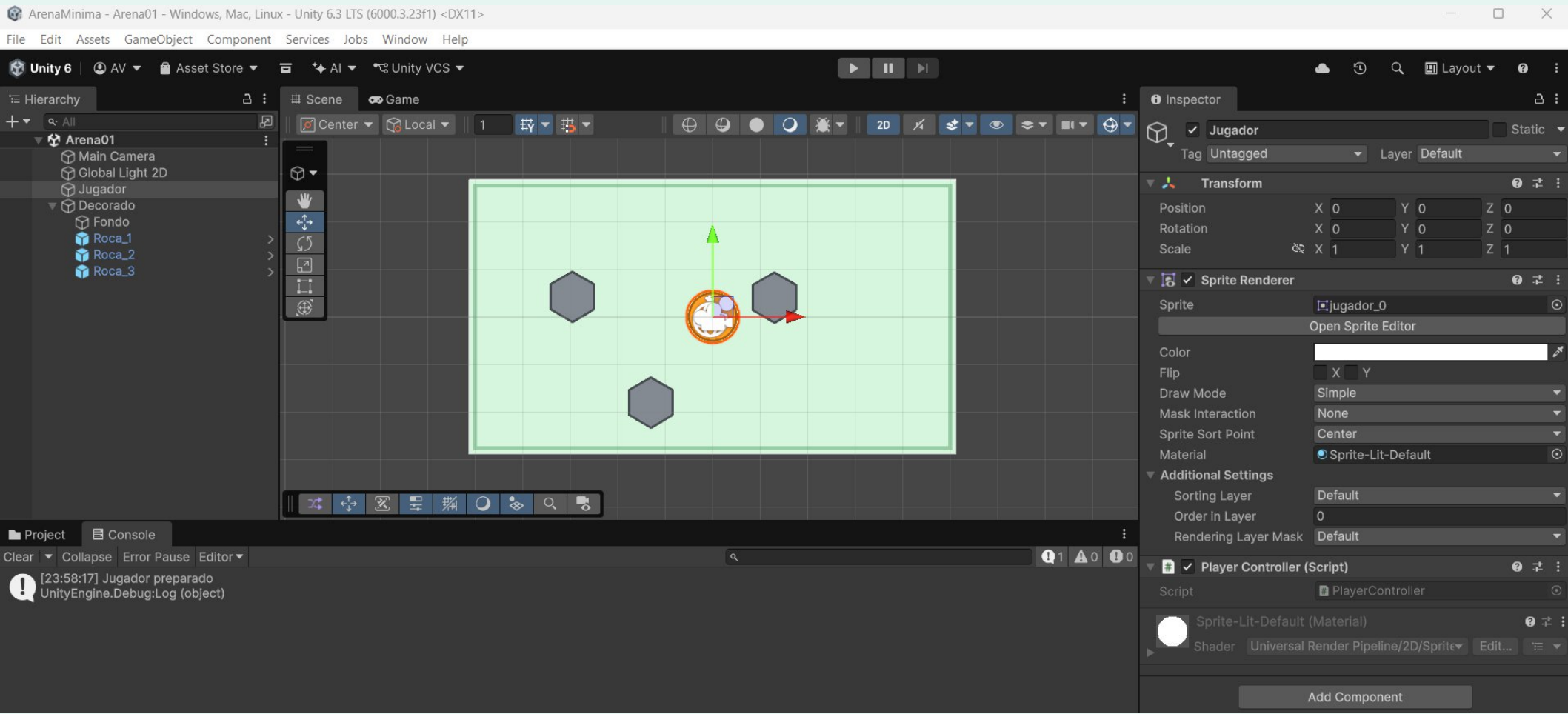
Primer script: comprobar que Unity lo ejecuta

- Editamos el código de PlayerController para poner “Debug.Log(“Jugador preparado”);” dentro de la función Start y para eliminar la función Update.
- Volvemos a Unity y esperamos a que compile.
- Le damos al Play, observamos Console y salimos del Play Mode.


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ArenaMinima
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features.
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Assembly-CSharp.csproj
PlayerController.cs
Assets > Scripts > PlayerController.cs
1 using UnityEngine;
2
3 public class PlayerController : MonoBehaviour
4 {
5 // Start is called once before the first ex
6 void Start()
7 {
8 Debug.Log("Jugador preparado");
9 }
10
11 }
12


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Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Sprite Renderer
Sprite jugador_0
Open Sprite Editor
Color
Flip X Y
Draw Mode Simple
Mask Interaction None
Sprite Sort Point Center
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Roca_3
DontDestroyOnLoad
Scene Game
Game Display 1 16:9 Aspect Scale 1.3x Play Focused Stats Gizmos
Project Console
Clear Collapse Error Pause Editor
[23:58:17] Jugador preparado
UnityEngine.Debug:Log (object)
Inspector
Jugador
Static
Tag Untagged Layer Default
Transform
Position X 0 Y 0 Z 0
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Sprite Renderer
Sprite jugador_0
Open Sprite Editor
Color
Flip X Y
Draw Mode Simple
Mask Interaction None
Sprite Sort Point Center
Material Sprite-Lit-Default
Additional Settings
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Player Controller (Script)
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Shader Universal Render Pipeline/2D/Sprite Edit...
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ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1) <DX11>
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Rotation X 0 Y 0 Z 0
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Mask Interaction None
Sprite Sort Point Center
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Additional Settings
Sorting Layer Default
Order in Layer 0
Rendering Layer Mask Default
Player Controller (Script)
Script PlayerController
Sprite-Lit-Default (Material)
Shader Universal Render Pipeline/2D/Sprite
Edit...
Add Component
Project Console
Clear Collapse Error Pause Editor
[23:58:17] Jugador preparado
UnityEngine.Debug:Log (object)

Configuración sin convertir todo en public

```
using UnityEngine;

public class PlayerController : MonoBehaviour
{
    [SerializeField] private float speed = 3f;
}
```

Inspector

Player Controller (Script)

Script PlayerController
Speed 3

Código `[SerializeField] private float speed = 3f;` →

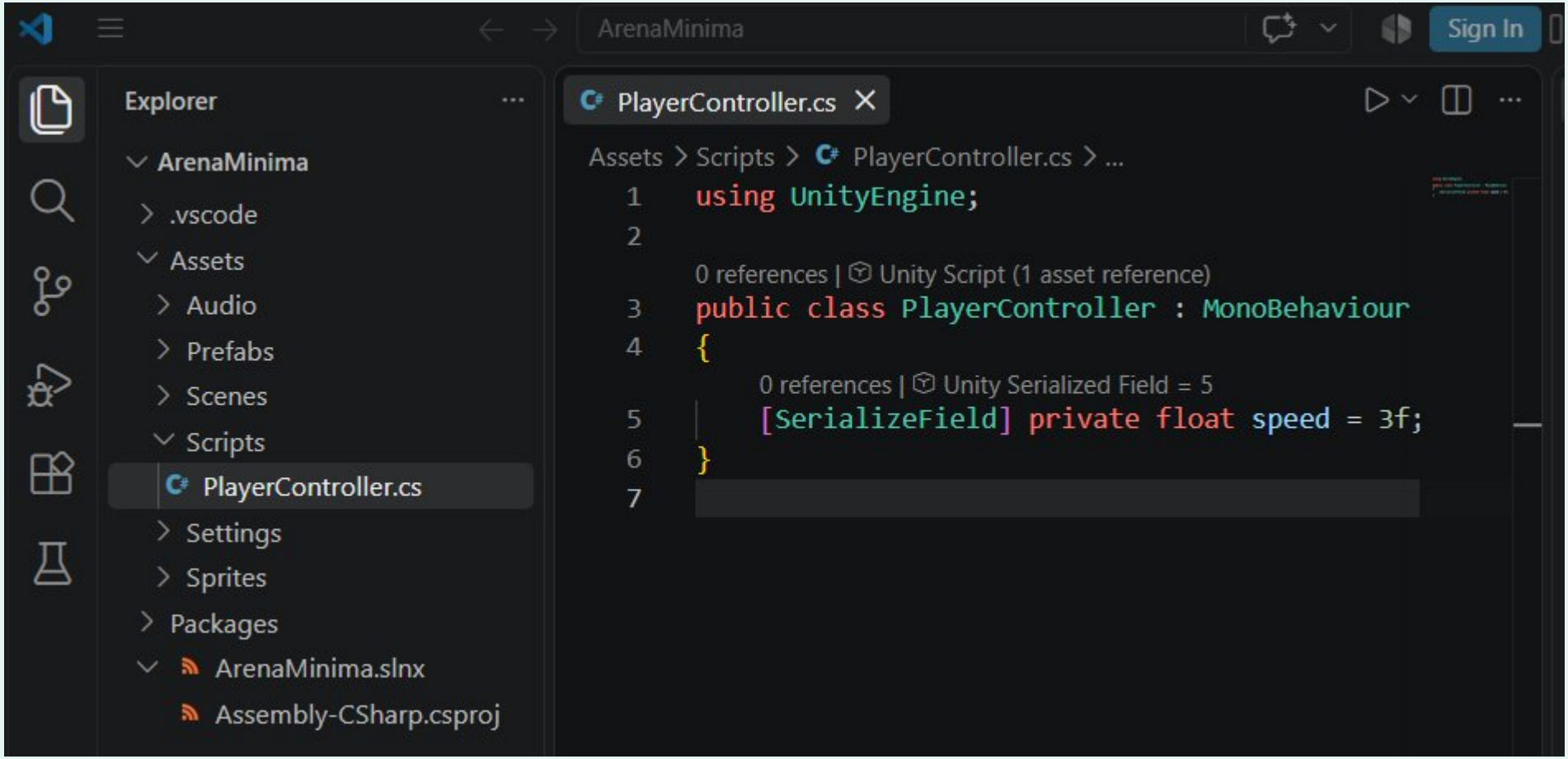
Configurable en Inspector
≠
public para todo el código

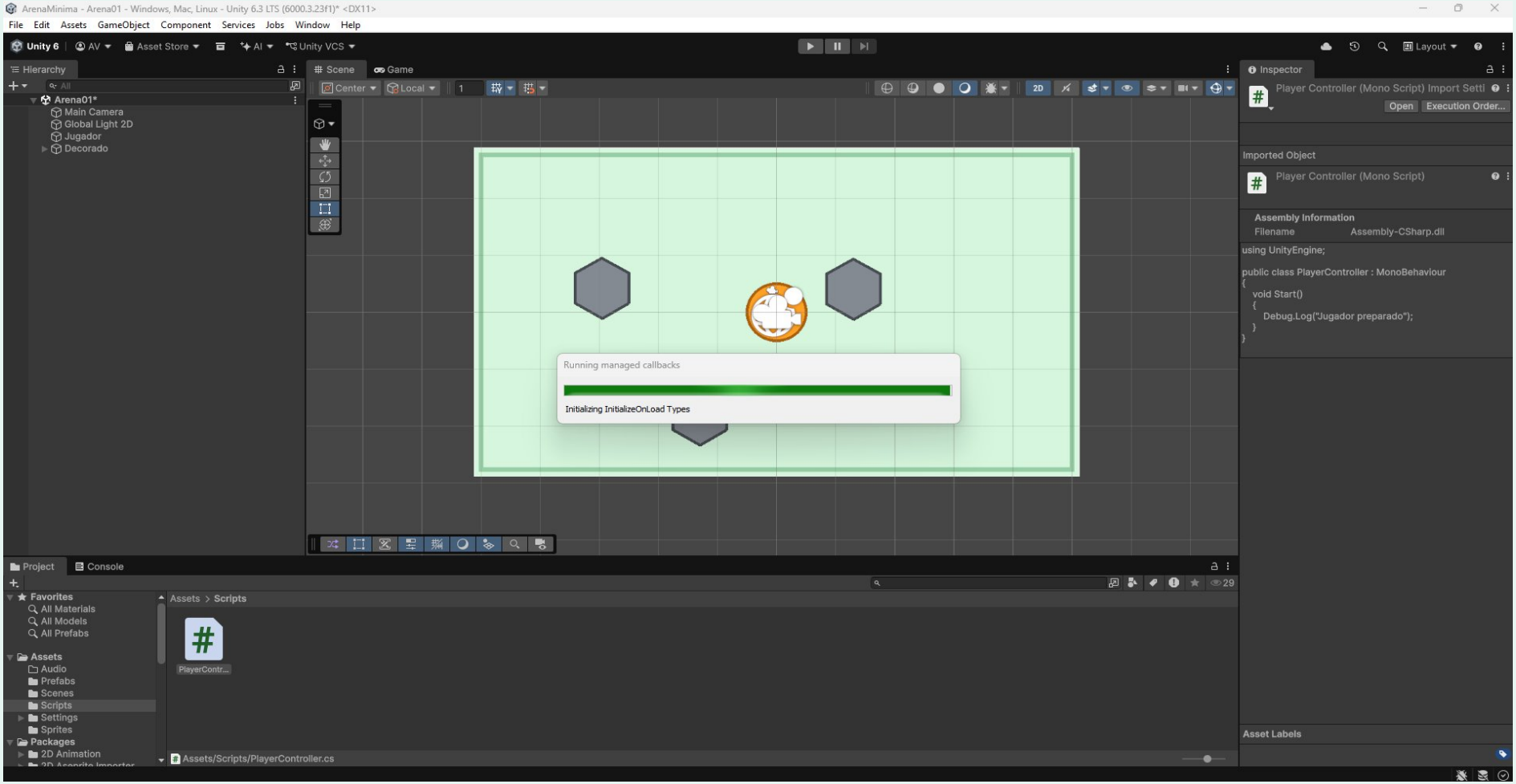
configuración → dato

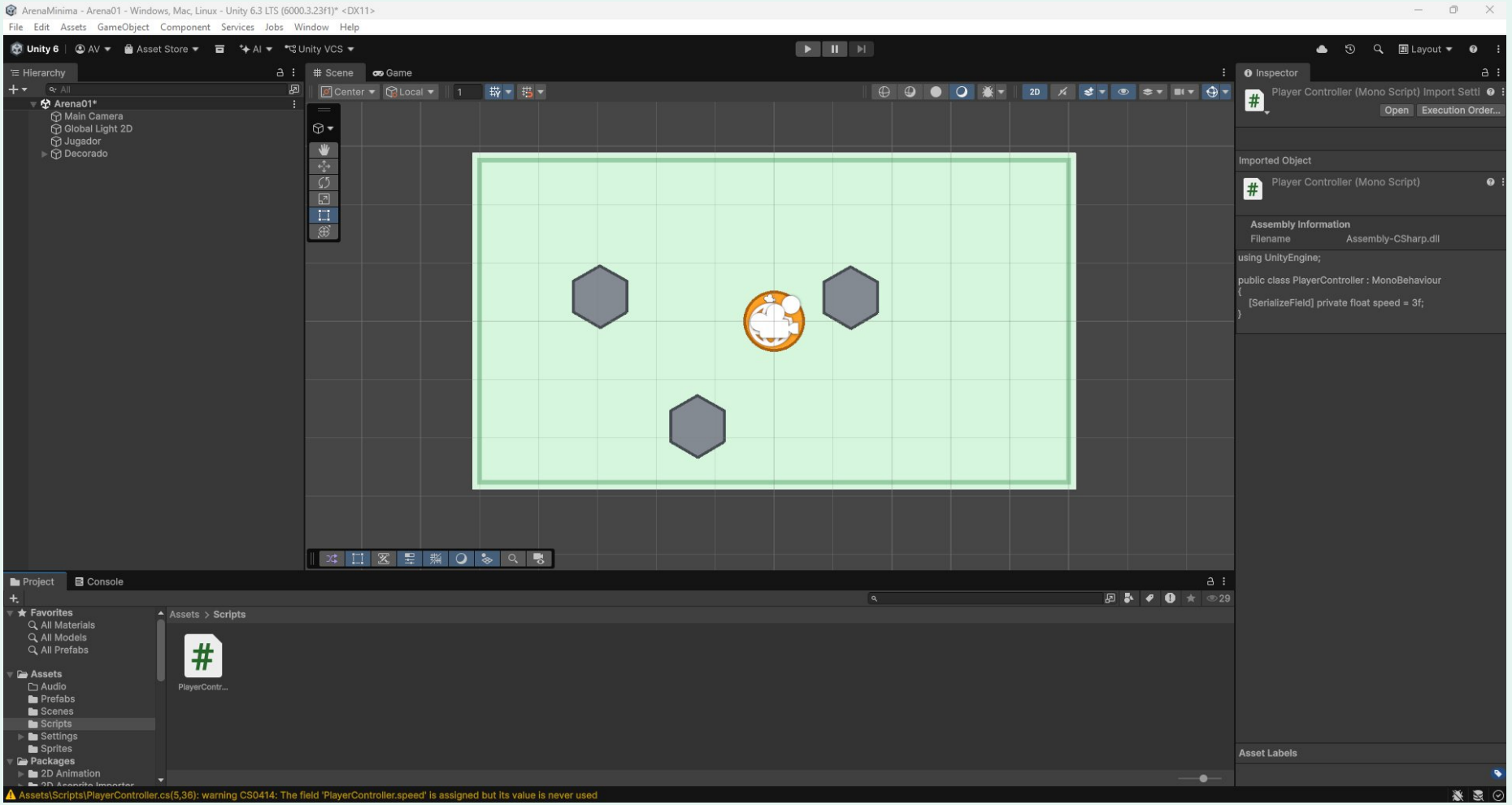
comportamiento → código

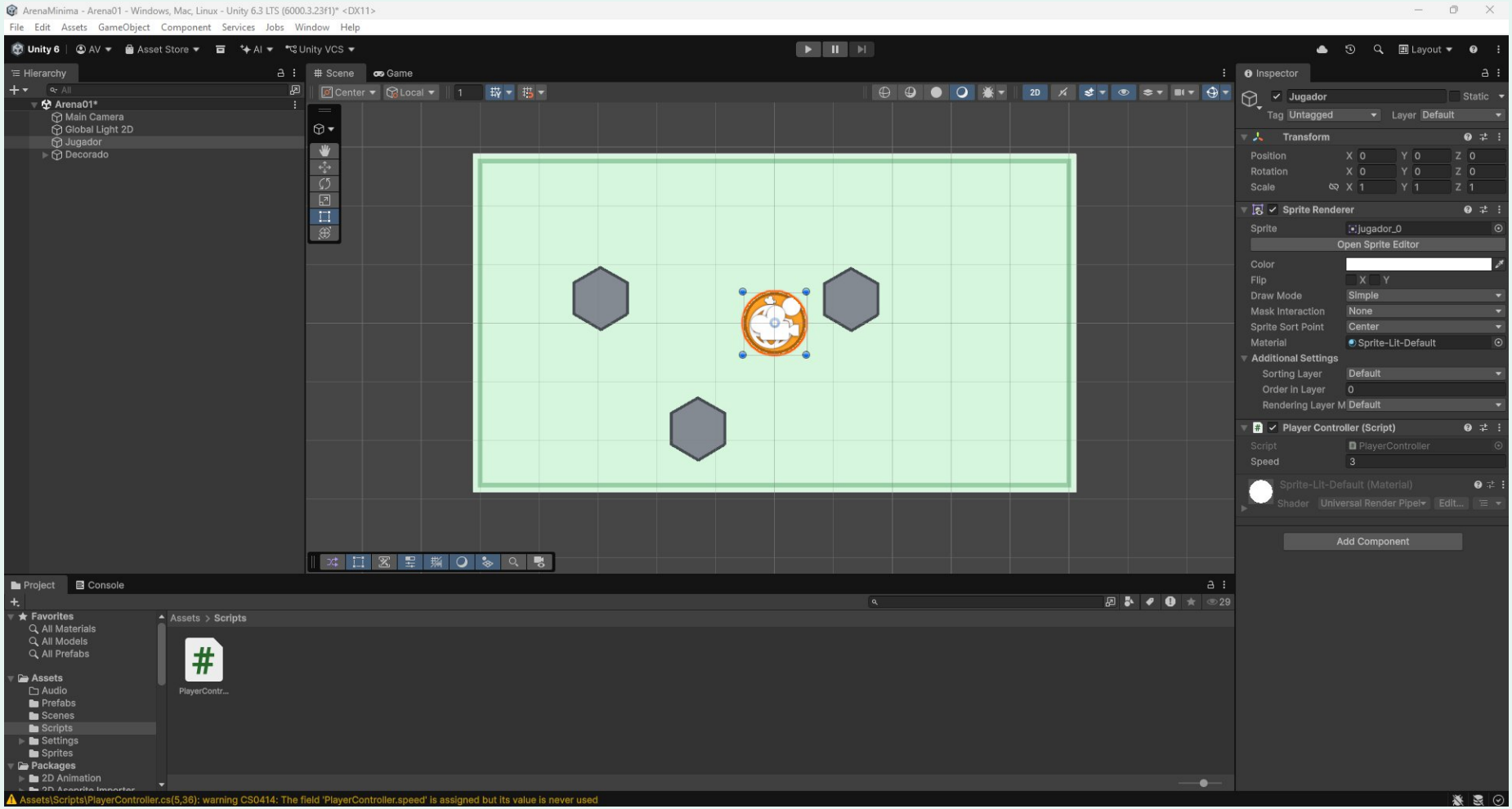
Configuración sin convertir todo en public

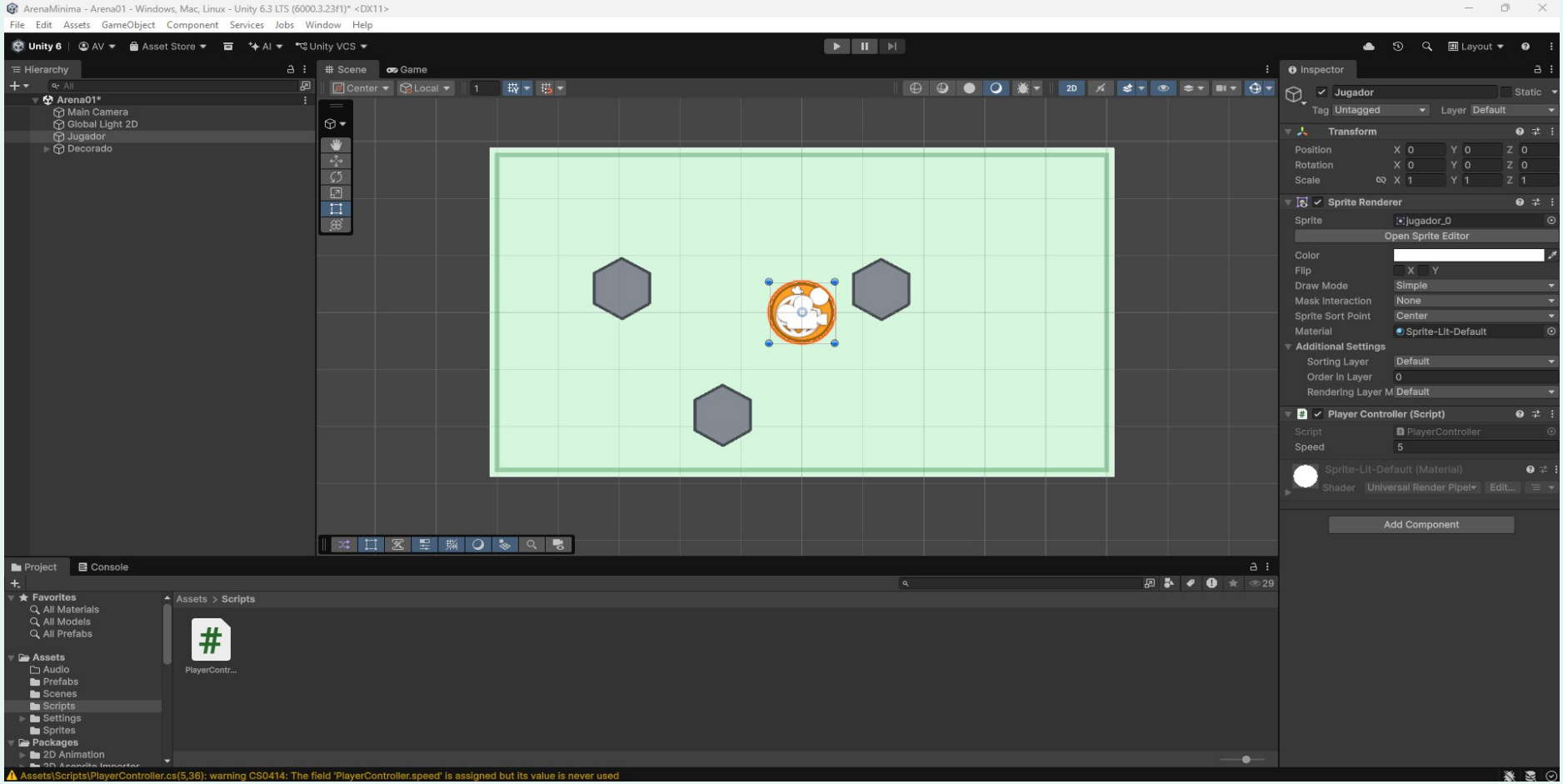
- Quitar el Debug.Log
- Añadir en la clase el campo: [SerializeField] private float speed = 3f;
- Guardar y esperar a que Unity compile.
- Seleccionar Jugador y ver que en Inspector aparece el campo speed con su valor editable.

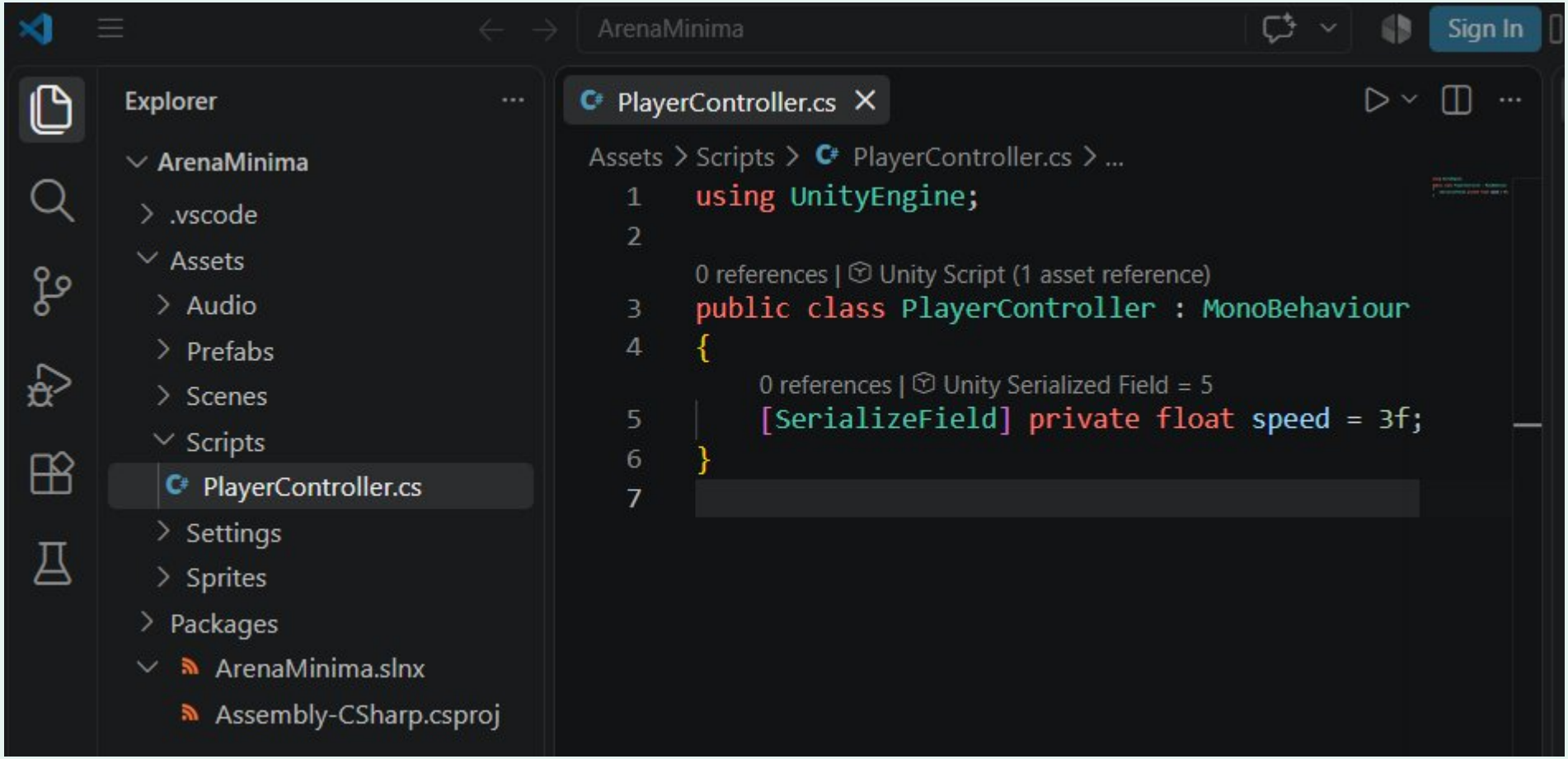

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ArenaMinima.slnx
Assembly-CSharp.csproj
PlayerController.cs
Assets > Scripts > PlayerController.cs > ...
1 using UnityEngine;
2
0 references | Unity Script (1 asset reference)
3 public class PlayerController : MonoBehaviour
4 {
0 references | Unity Serialized Field = 5
5 [SerializeField] private float speed = 3f;
6 }
7


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1)* <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Unity 6
Hierarchy
Arena01*
Main Camera
Global Light 2D
Jugador
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Scene
Game
Running managed callbacks
Initializing InitializeOnLoad Types
Inspector
Player Controller (Mono Script) Import Setti
Open
Execution Order...
Imported Object
Player Controller (Mono Script)
Assembly Information
Filename Assembly-CSharp.dll
using UnityEngine;
public class PlayerController : MonoBehaviour
{
void Start()
{
Debug.Log("Jugador preparado");
}
}
Project
Console
Favorites
All Materials
All Models
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Assets > Scripts
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Assets/Scripts/PlayerController.cs
Asset Labels


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Inspector
Player Controller (Mono Script) Import Setti
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Execution Order...
Imported Object
Player Controller (Mono Script)
Assembly Information
Filename Assembly-CSharp.dll
using UnityEngine;
public class PlayerController : MonoBehaviour
{
[SerializeField] private float speed = 3f;
}
Project
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Assets > Scripts
PlayerContr...
Asset Labels
Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used

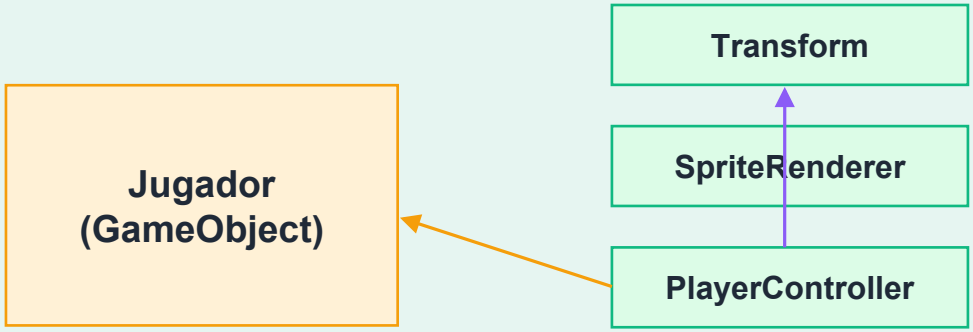

ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1)* <DX11>
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Tag Untagged
Layer Default
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Scale
Sprite Renderer
Sprite
jugador_0
Open Sprite Editor
Color
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Draw Mode
Simple
Mask Interaction
None
Sprite Sort Point
Center
Material
Sprite-Lit-Default
Additional Settings
Sorting Layer
Default
Order in Layer
Rendering Layer M
Player Controller (Script)
Script
PlayerController
Speed
3
Sprite-Lit-Default (Material)
Shader
Universal Render Pipel
Edit...
Add Component
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Assets > Scripts
PlayerContr...
Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1)* <DX11>
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Sprite jugador_0
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Mask Interaction None
Sprite Sort Point Center
Material Sprite-Lit-Default
Additional Settings
Sorting Layer Default
Order in Layer 0
Rendering Layer M Default
Player Controller (Script)
Script PlayerController
Speed 5
Sprite-Lit-Default (Material)
Shader Universal Render Pipel
Edit...
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Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used


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PlayerController.cs
Assets > Scripts > PlayerController.cs > ...
1 using UnityEngine;
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0 references | Unity Script (1 asset reference)
3 public class PlayerController : MonoBehaviour
4 {
0 references | Unity Serialized Field = 5
5 [SerializeField] private float speed = 3f;
6 }
7

Referencias: trabajar con lo que ya existe

`PlayerController` está en `Jugador`



`gameObject` → referencia al GameObject `Jugador`

`transform` → referencia a su Component `Transform`

Por ejemplo:

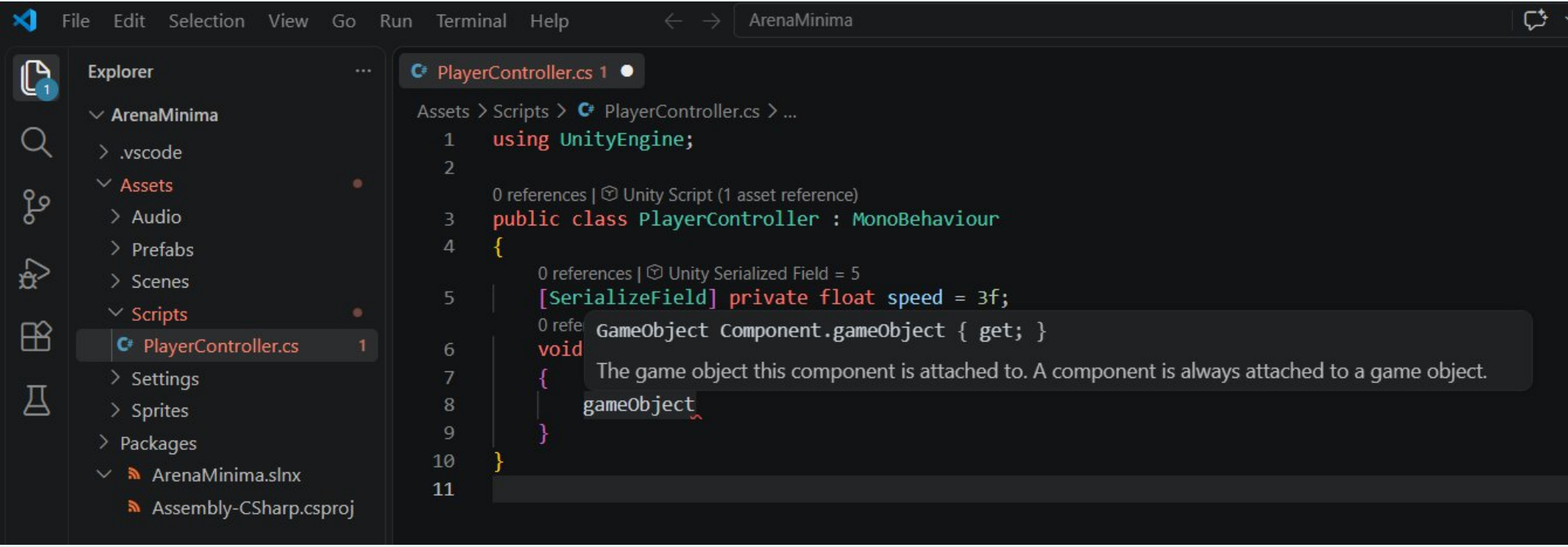
```
gameObject.name
transform.position
```

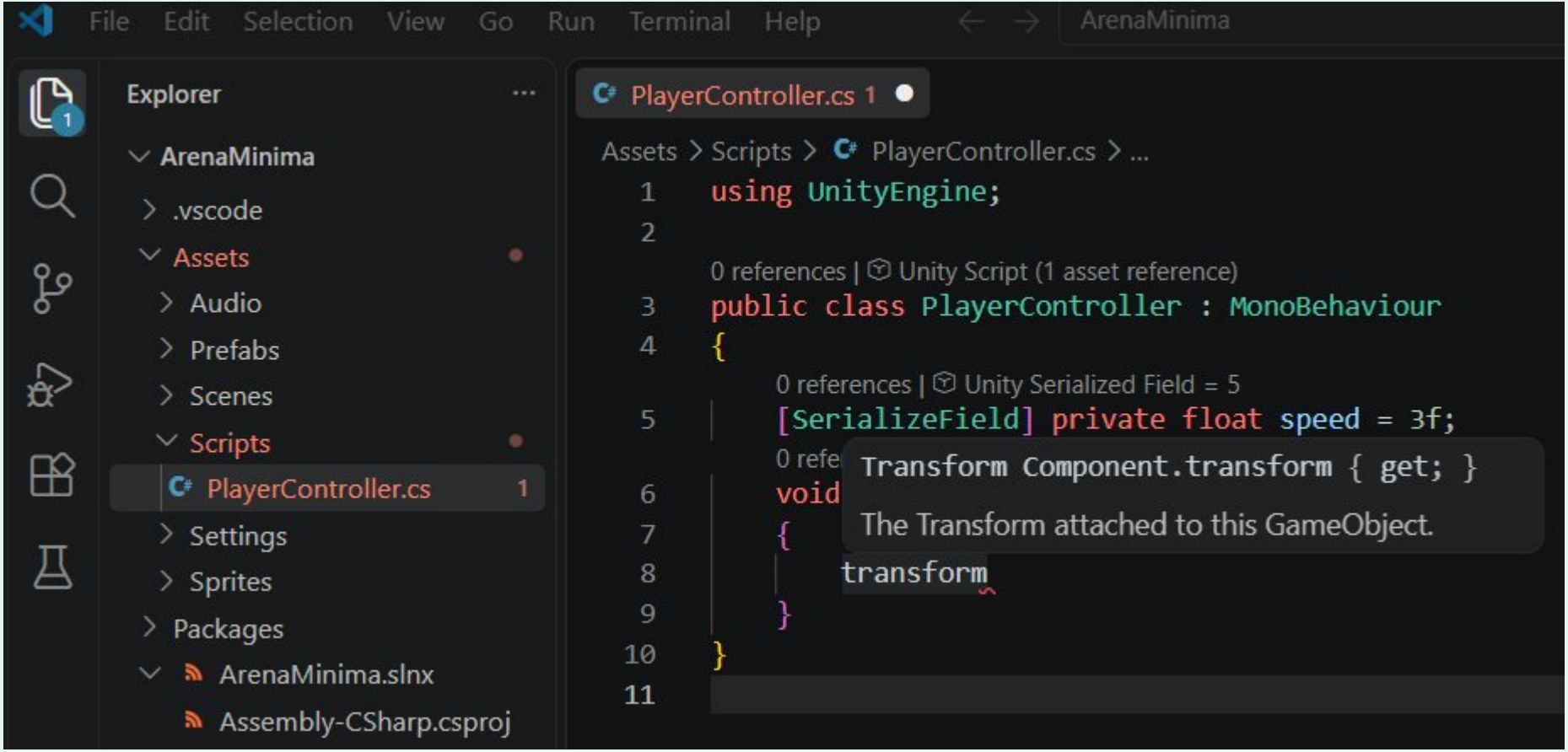
referencia ≠ copia

Tener una referencia ≠ crear otro objeto o Component

Referencias: trabajar con lo que ya existe

- Desde VS Code, en el script, escribir el método void Start() {}
- Probar a escribir dentro del método gameObject y dejar el cursor sobre ese texto para ver el panel contextual.
- Probar a escribir dentro del método transform y dejar el cursor sobre ese texto para ver el panel contextual.


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ArenaMinima.slnx
Assembly-CSharp.csproj
PlayerController.cs 1
Assets > Scripts > PlayerController.cs > ...
using UnityEngine;
0 references | Unity Script (1 asset reference)
public class PlayerController : MonoBehaviour
{
0 references | Unity Serialized Field = 5
[SerializeField] private float speed = 3f;
GameObject Component.gameObject { get; }
The game object this component is attached to. A component is always attached to a game object.
gameObject
}
}


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ArenaMinima.slnx
Assembly-CSharp.csproj
PlayerController.cs 1
Assets > Scripts > PlayerController.cs > ...
using UnityEngine;
0 references | Unity Script (1 asset reference)
public class PlayerController : MonoBehaviour
{
0 references | Unity Serialized Field = 5
[SerializeField] private float speed = 3f;
0 refe
void
{
transform
}
}
Transform Component.transform { get; }
The Transform attached to this GameObject.

Unity decide cuándo llama a nuestro código

Unity controla el ciclo de ejecución

Inicio

Start()
una vez antes del primer Update cuando el Component está habilitado

Frames renderizados



Ciclo de actualización fija



puede haber 0, 1 o varias llamadas entre frames

En esta presentación: movimiento programado → Update

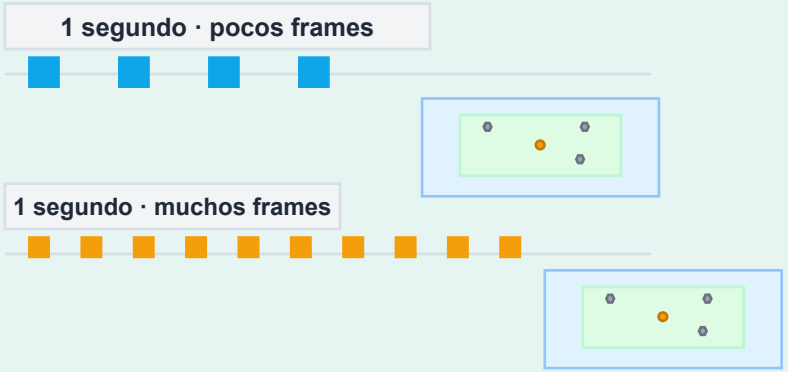
En la próxima presentación: física → reutilizaremos FixedUpdate

`Start` ≠ constructor
`Update` ≠ `while(true)` escrito por nosotros

Mover por tiempo, no por cantidad de frames

Problema

distancia fija por frame
↓
más frames → más distancia en el mismo segundo
↓
velocidad dependiente del framerate



Para nuestro prototipo

dirección × velocidad × tiempo = desplazamiento

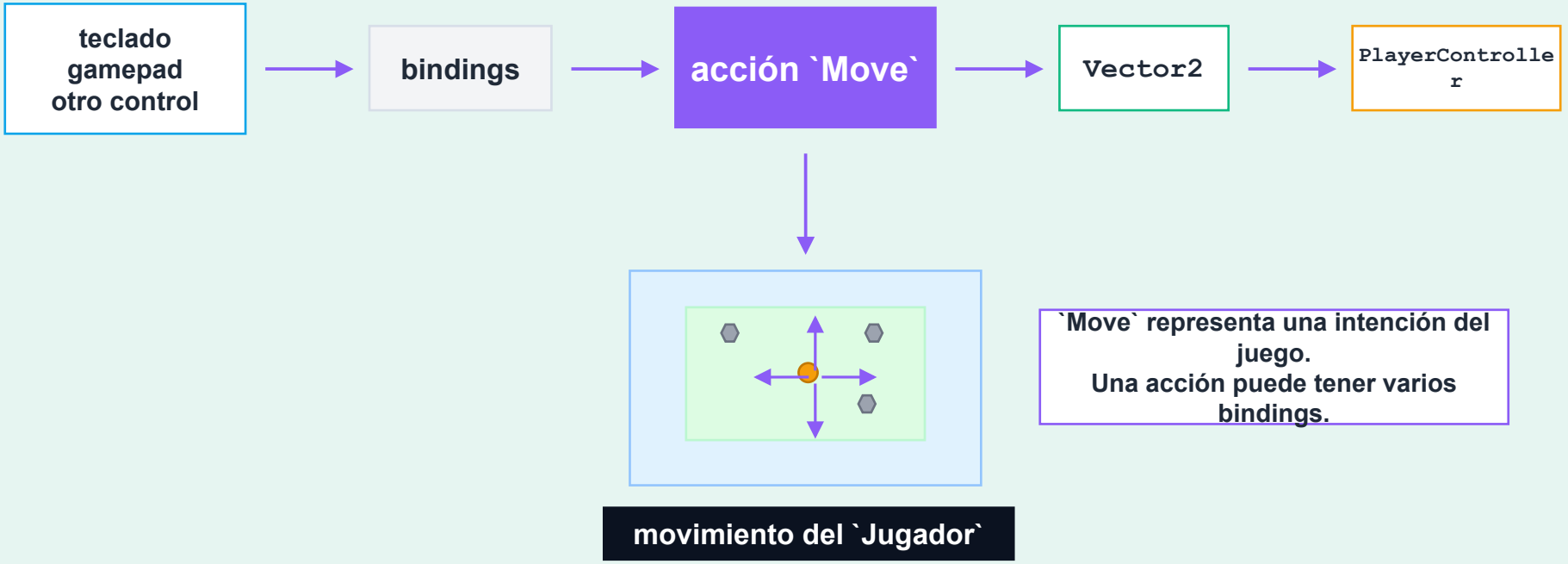
En Unity:
`Time.deltaTime`
= segundos transcurridos desde el frame anterior

```
dirección × speed × Time.deltaTime
```

velocidad × tiempo → desplazamiento

Entrada: una acción no es una tecla

control físico ≠ acción lógica



Configuración mínima de Move

Input Actions

Action Map:
`Player`

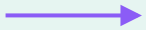
Action:
`Move`

Action Type:
`Value`

valor esperado:
`Vector2`

Binding:
`2D Vector`

Up → `W`
Down → `S`
Left → `A`
Right → `D`



GameObject `Jugador`

PlayerInput

Actions → nuestro asset
Default Map → `Player`
Behavior → `Send Messages`

Component

PlayerInput

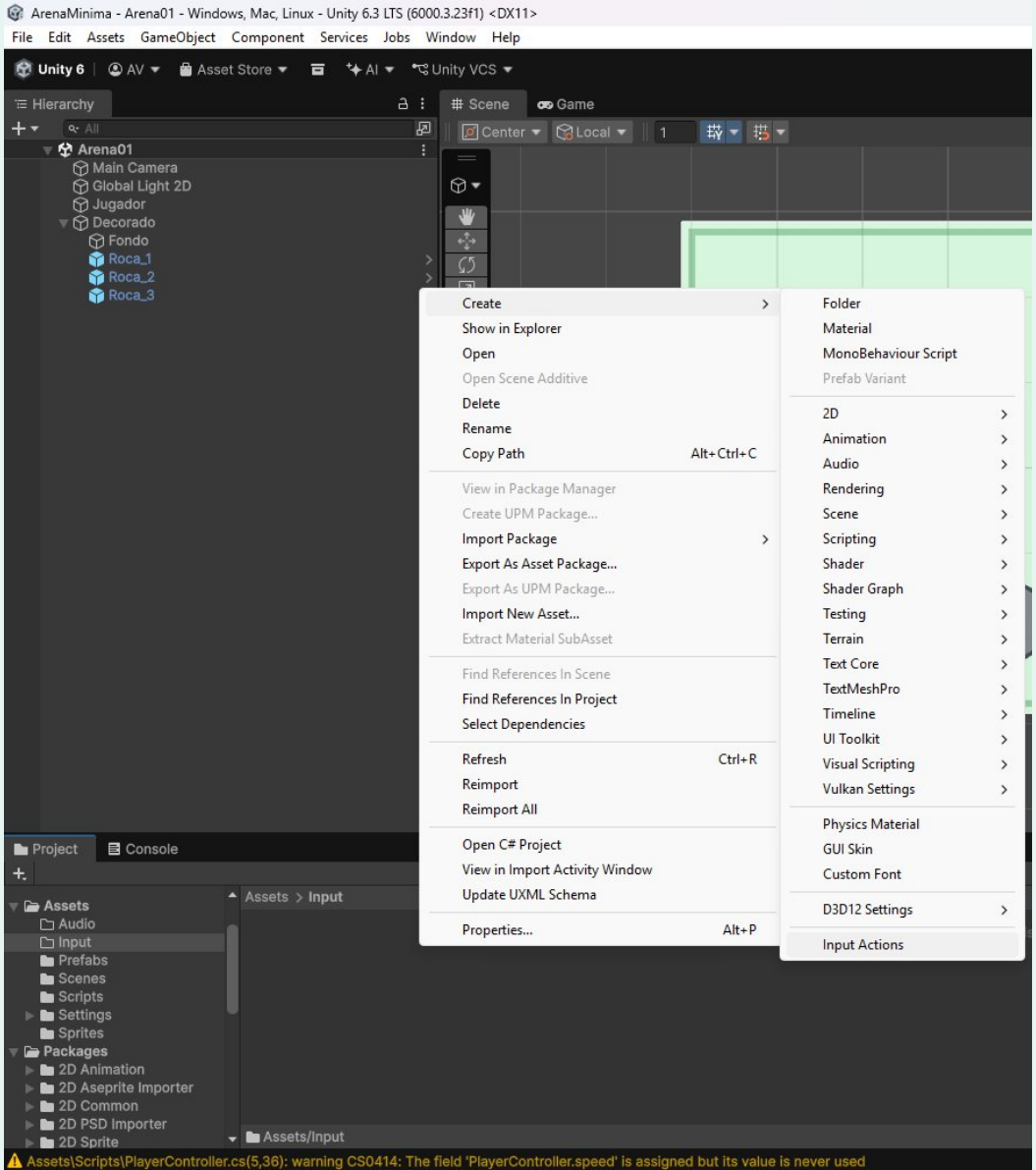
Default Map
`Player`

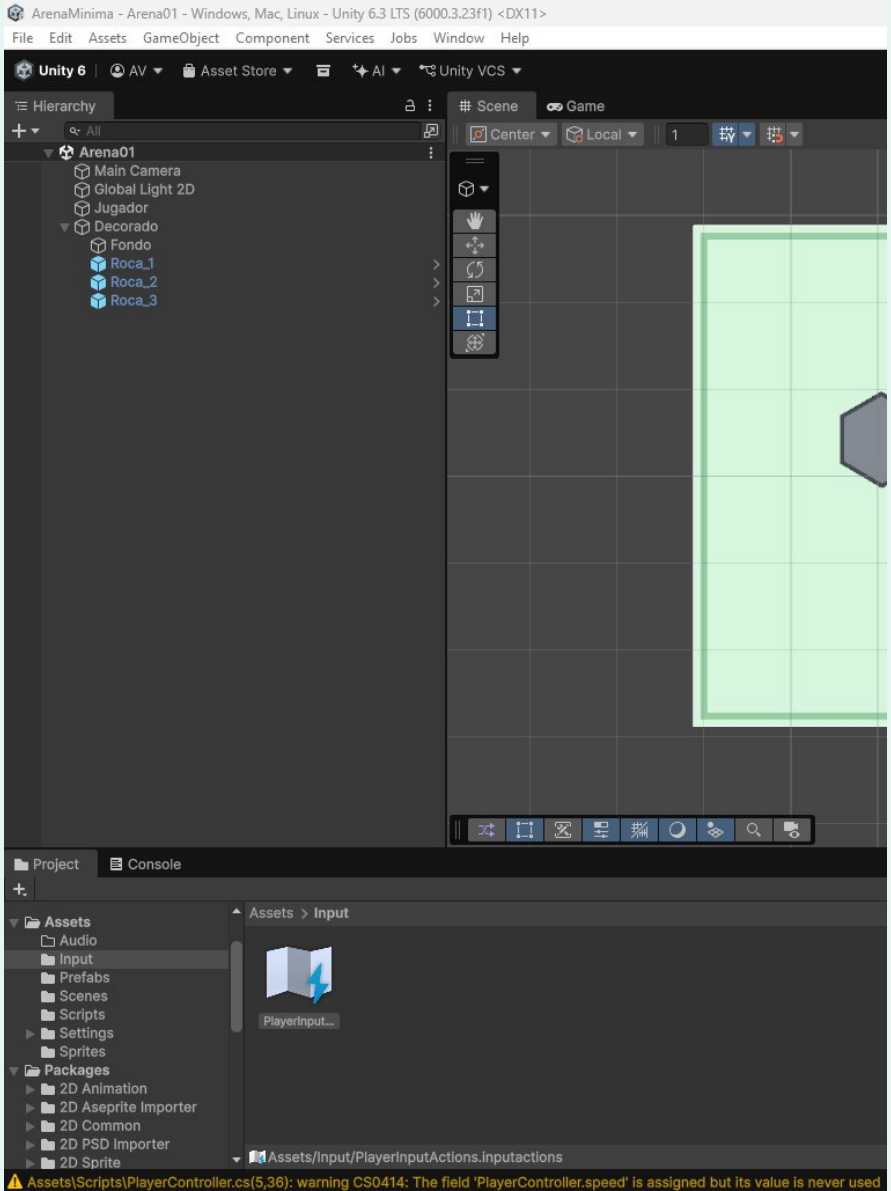
Precondición de clase:
Input System ya está instalado y operativo.

Una sola ruta completa:
`Move` → `PlayerInput` → `OnMove`

Configuración mínima de Move

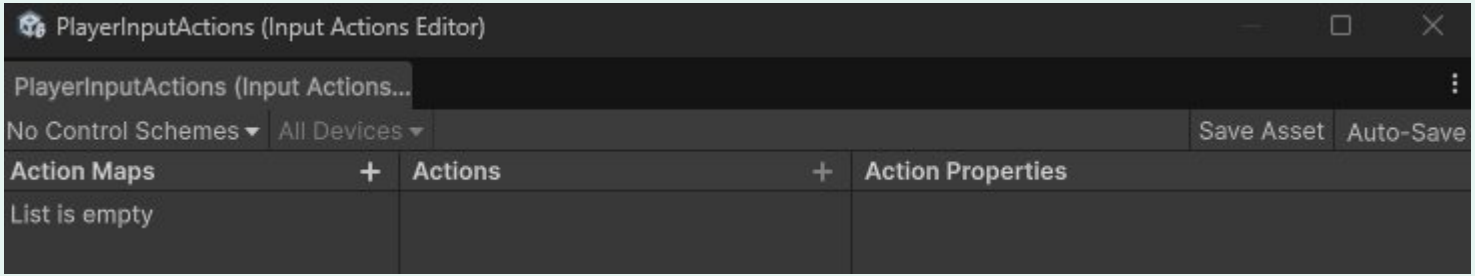
- En Unity, en Project, crear la carpeta Assets/Input.
- Click derecho en el espacio en blanco de la carpeta Assets/Input, vía el panel contextual crear un asset de Input Actions (Create -> Input Actions) y llamarlo "PlayerInputActions".

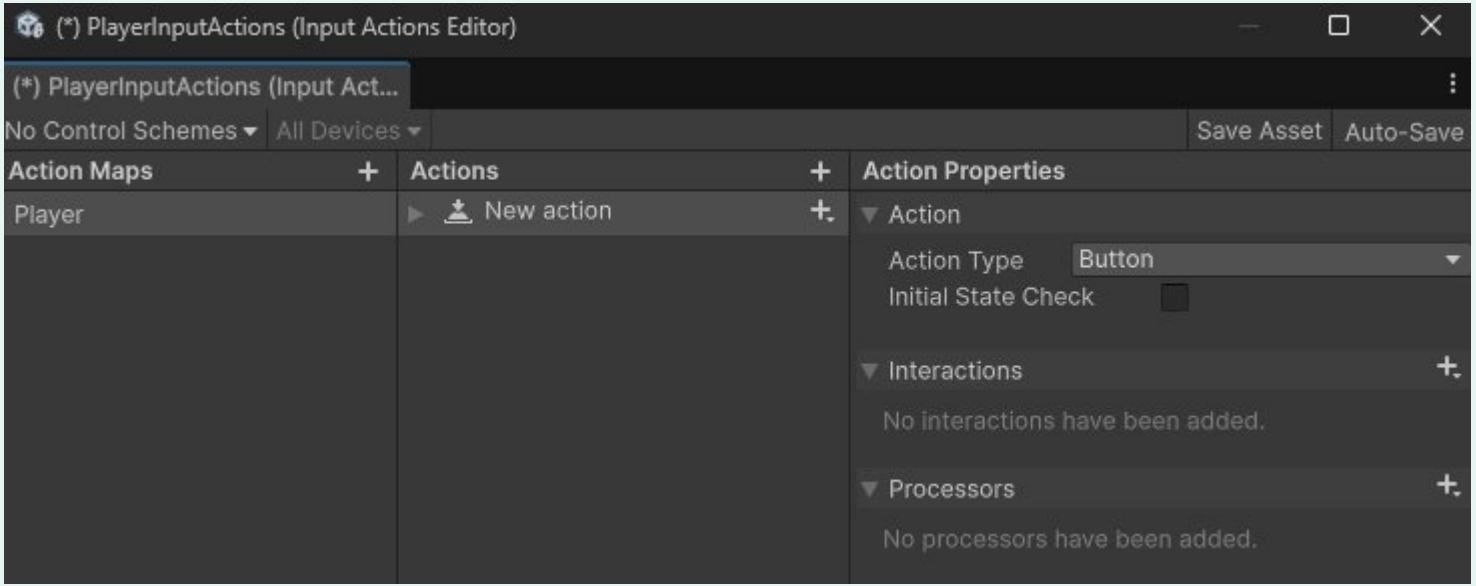

ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1) <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
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Arena01
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Global Light 2D
Jugador
Decorado
Fondo
Roca_1
Roca_2
Roca_3
Create
Show in Explorer
Open
Open Scene Additive
Delete
Rename
Copy Path Alt+Ctrl+C
View in Package Manager
Create UPM Package...
Import Package
Export As Asset Package...
Export As UPM Package...
Import New Asset...
Extract Material SubAsset
Find References In Scene
Find References In Project
Select Dependencies
Refresh Ctrl+R
Reimport
Reimport All
Open C# Project
View in Import Activity Window
Update UXML Schema
Properties... Alt+P
Folder
Material
MonoBehaviour Script
Prefab Variant
2D
Animation
Audio
Rendering
Scene
Scripting
Shader
Shader Graph
Testing
Terrain
Text Core
TextMeshPro
Timeline
UI Toolkit
Visual Scripting
Vulkan Settings
Physics Material
GUI Skin
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Packages
2D Animation
2D Aseprite Importer
2D Common
2D PSD Importer
2D Sprite
Assets > Input
Assets/Input
Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used

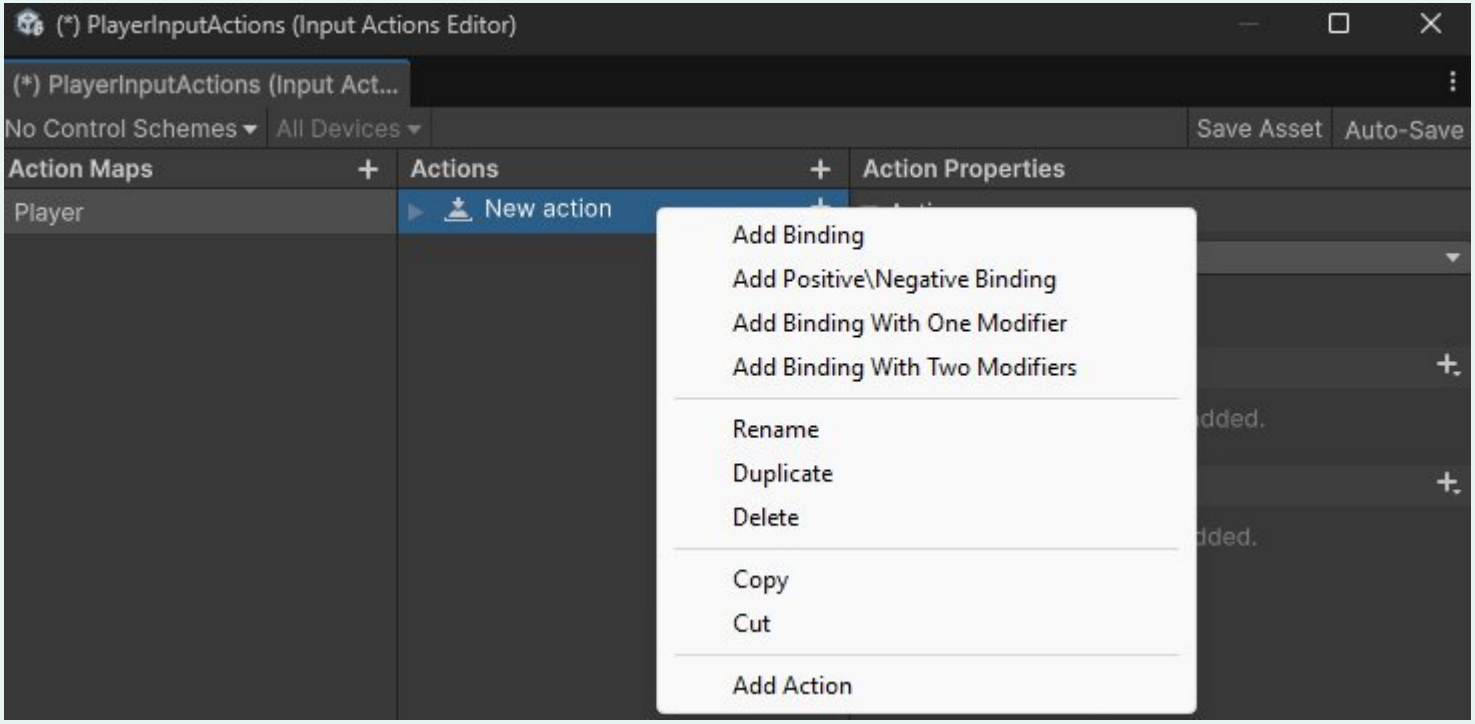

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2D Animation
2D Aseprite Importer
2D Common
2D PSD Importer
2D Sprite
Assets > Input
PlayerInput...
Assets/Input/PlayerInputActions.inputactions
Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used

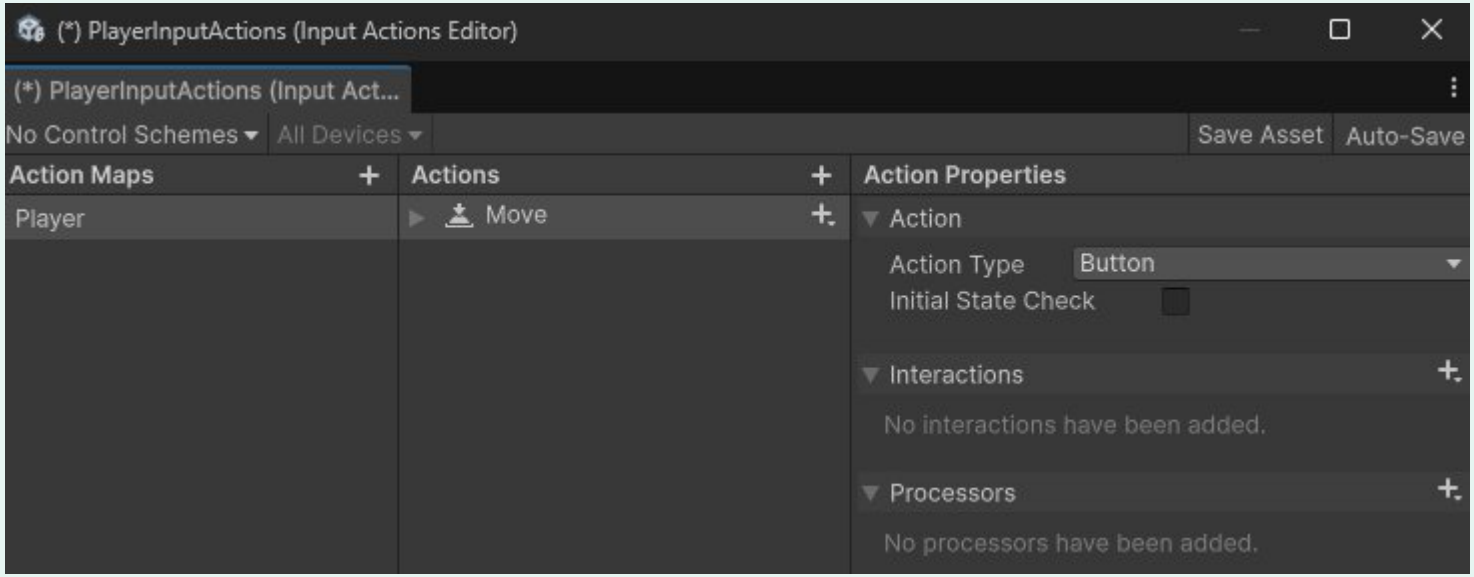
Configuración mínima de Move

- En Unity, en Project, hacer doble click sobre ese asset para abrir el editor de Input Actions.
- En Unity, en PlayerInputActions, click izquierdo en el "+" del panel de Action Maps y darle de nombre "Player".
- En el Action Map "Player", con click derecho renombrar la acción "New action" a "Move" (o crear una nueva).


PlayerInputActions (Input Actions Editor)
PlayerInputActions (Input Actions...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Actions
Action Properties
List is empty

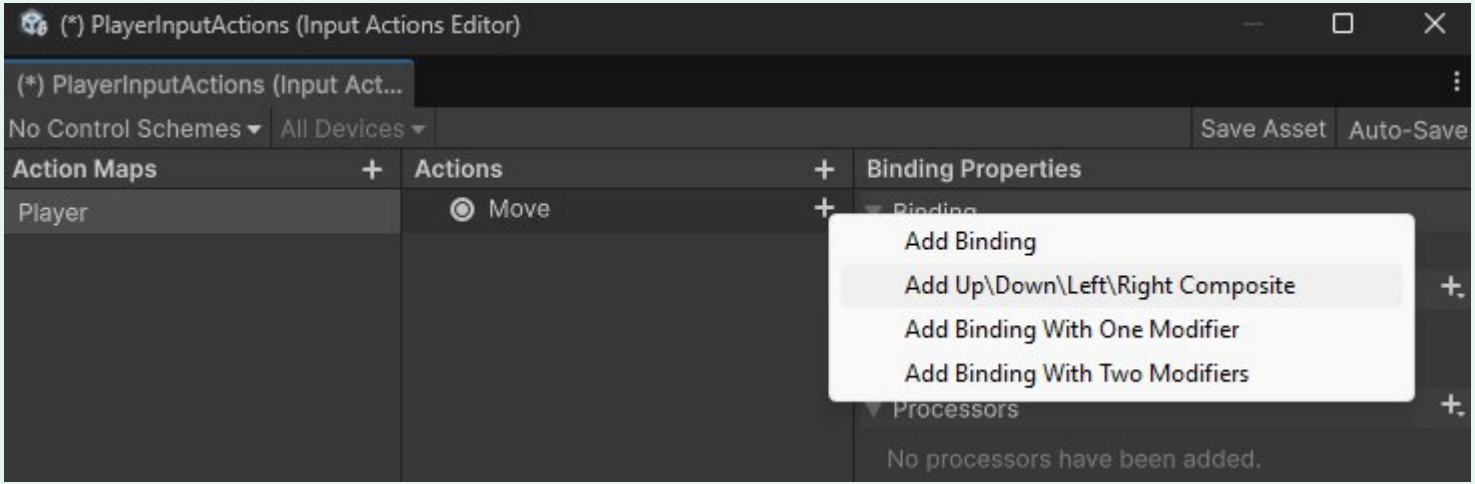

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Actions
Action Properties
Player
New action
Action
Action Type
Button
Initial State Check
Interactions
No interactions have been added.
Processors
No processors have been added.

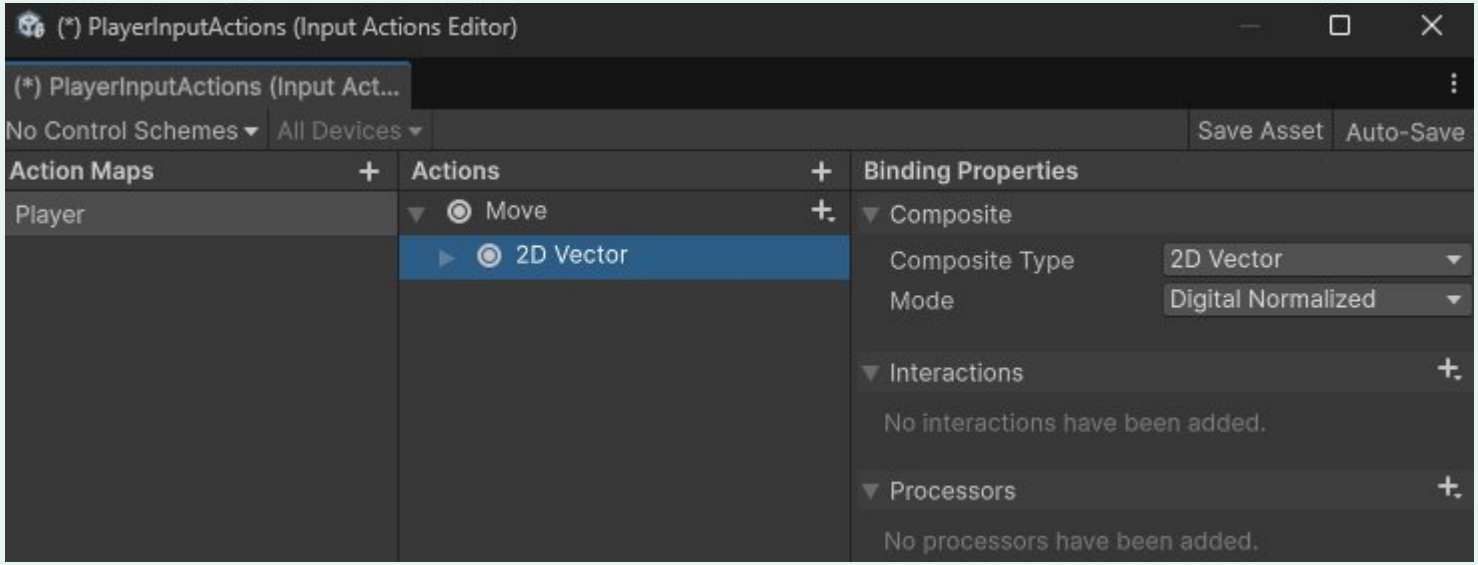

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Actions
Action Properties
Player
New action
Add Binding
Add Positive\Negative Binding
Add Binding With One Modifier
Add Binding With Two Modifiers
Rename
Duplicate
Delete
Copy
Cut
Add Action


(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
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Action Properties
Player
Move
Action
Action Type
Button
Initial State Check
Interactions
No interactions have been added.
Processors
No processors have been added.

Configuración mínima de Move

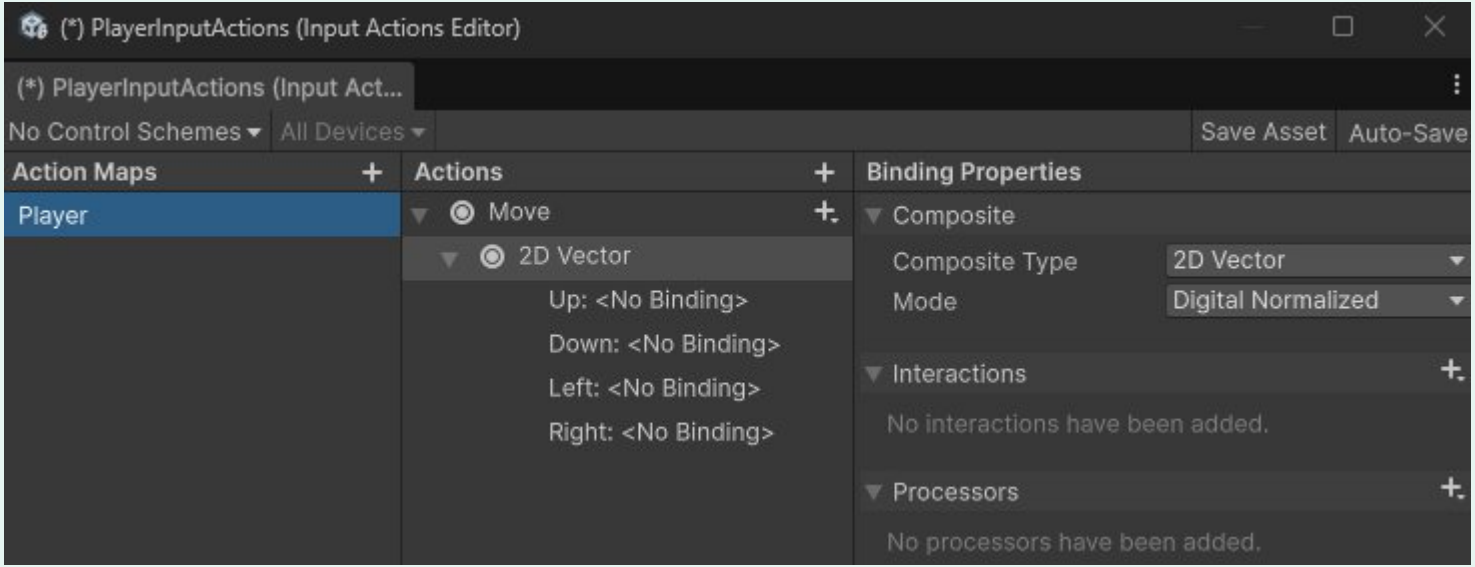
- En Unity, en PlayerInputActions, en el Action Map "Player", en el Action "Move", configurar "Action Type" -> "Value" y "Control Type" -> "Vector 2".
- Click izquierdo en el "+" asociado a la acción “Move”y click izquierdo en "Add Up\Down\Left\Right Composite" para añadir un binding.

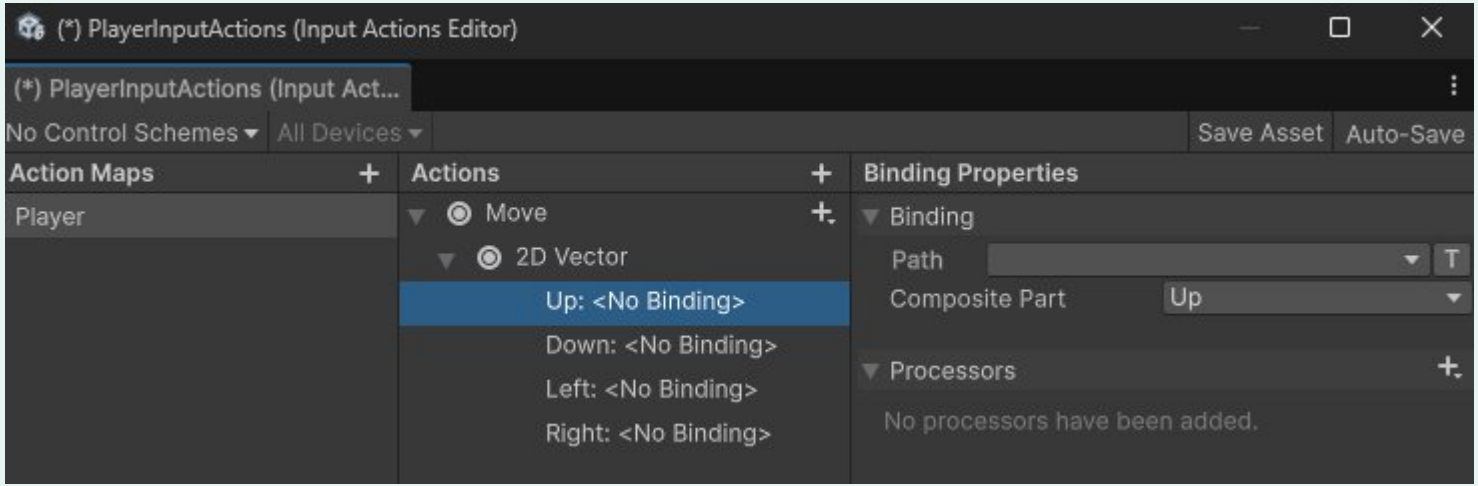

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
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Action Maps
Actions
Binding Properties
Player
Move
Add Binding
Add Up\Down\Left\Right Composite
Add Binding With One Modifier
Add Binding With Two Modifiers
Processors
No processors have been added.

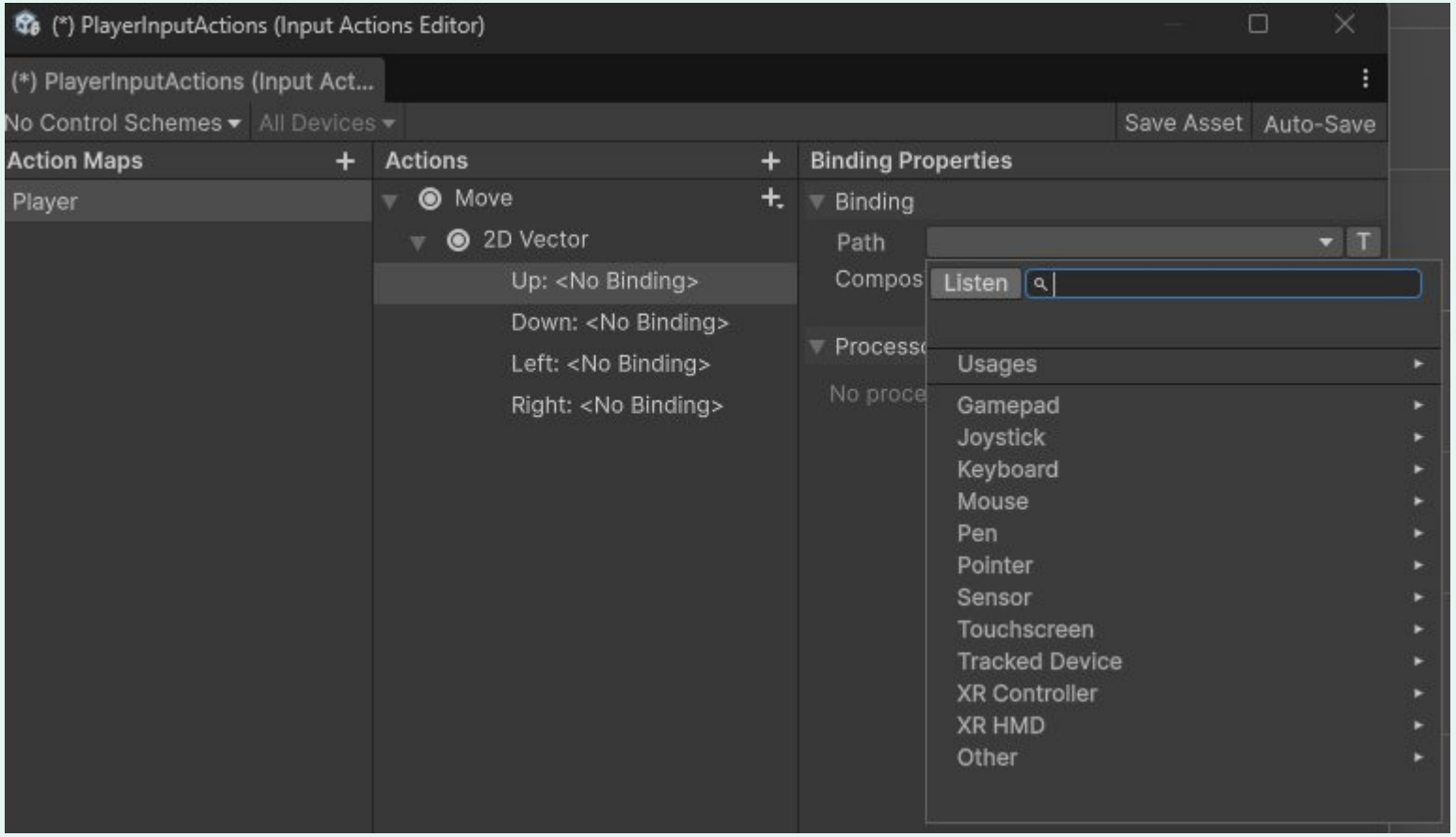

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Player
Actions
Move
2D Vector
Binding Properties
Composite
Composite Type
2D Vector
Mode
Digital Normalized
Interactions
No interactions have been added.
Processors
No processors have been added.

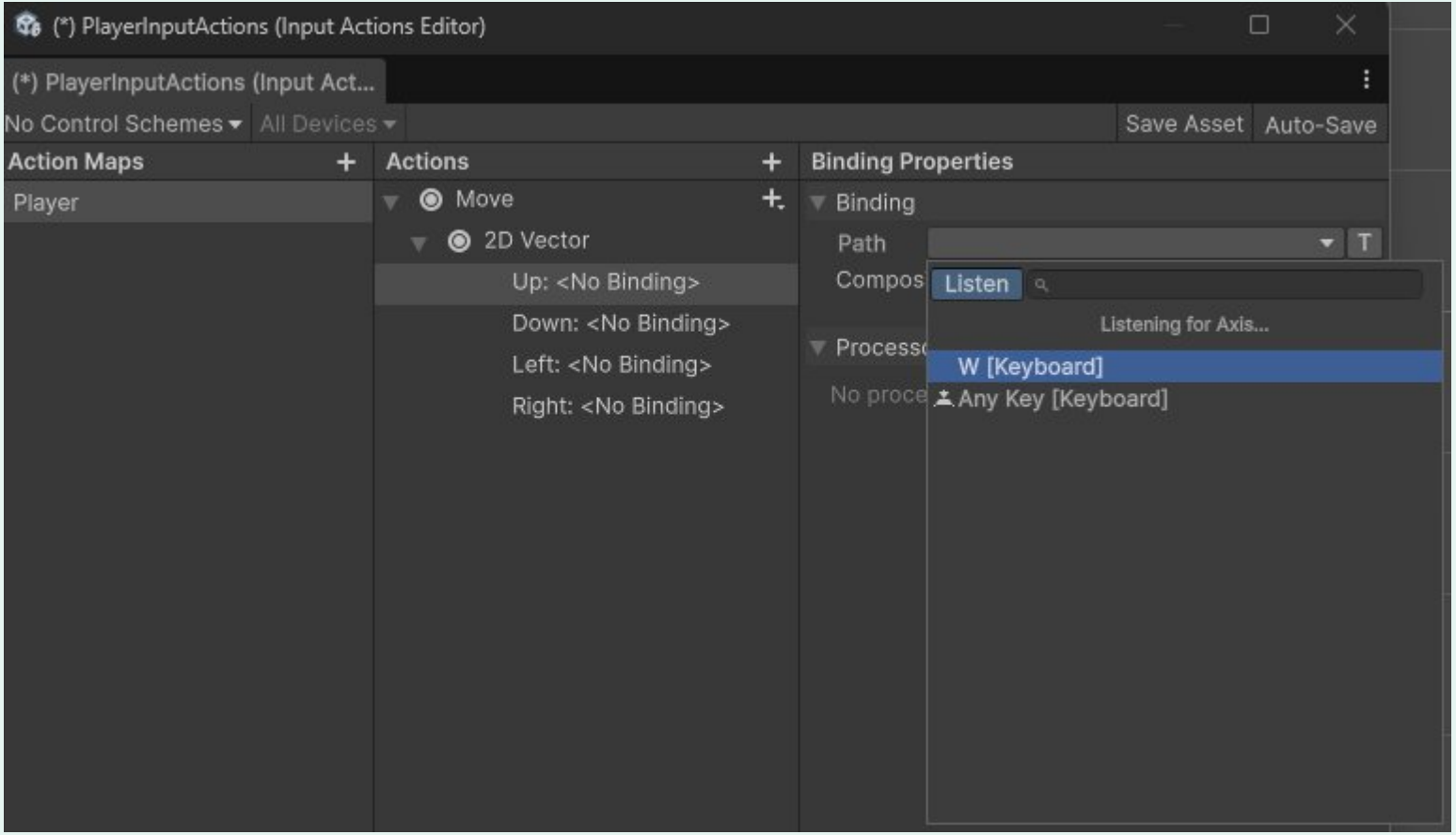
Configuración mínima de Move

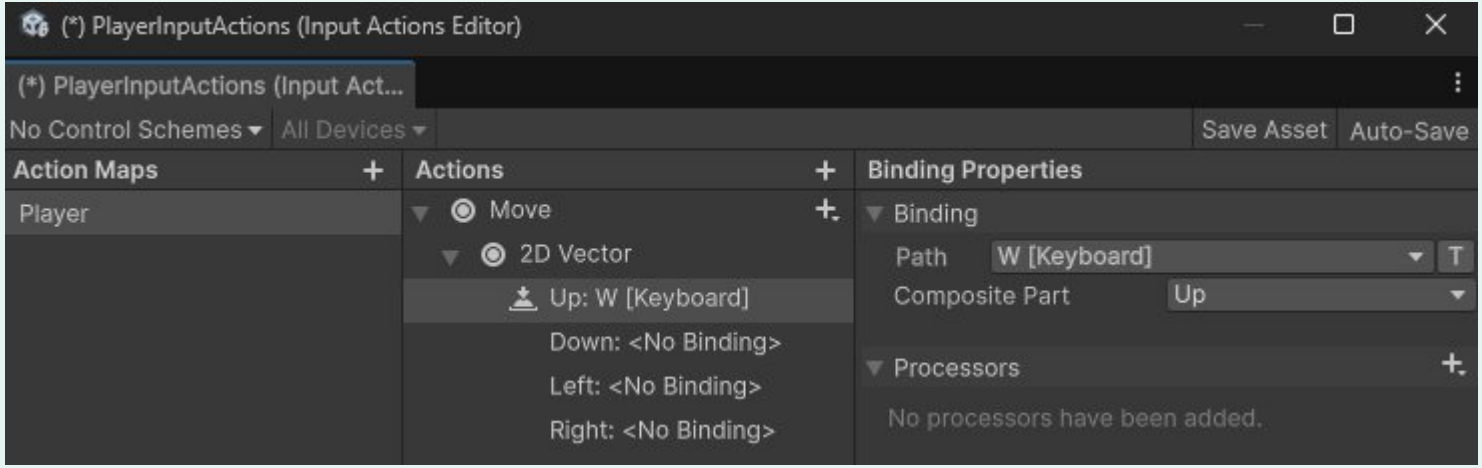
- En Unity, en PlayerInputActions, en el Action Map "Player", en el Action "Move", desplegar "2D Vector".
- En 2D Vector, caso por caso hacer click izquierdo en el caso, click izquierdo en el desplegable de "Path", click izquierdo en "Listen" y pulsar la tecla correspondiente (WSAD).

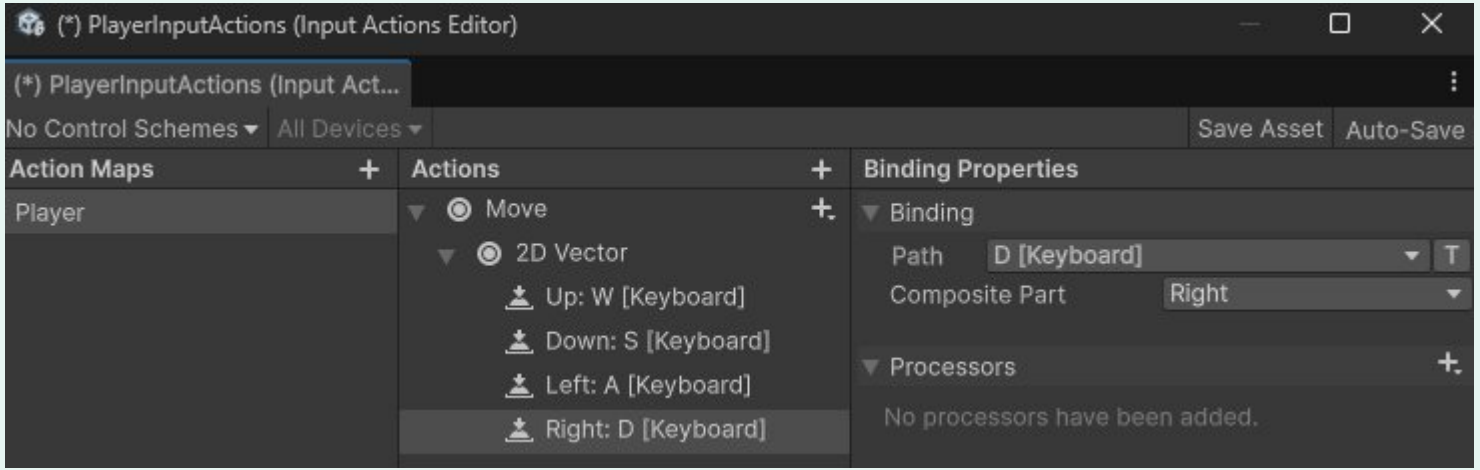

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
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Player
Actions
Move
2D Vector
Up: <No Binding>
Down: <No Binding>
Left: <No Binding>
Right: <No Binding>
Binding Properties
Composite
Composite Type
2D Vector
Mode
Digital Normalized
Interactions
No interactions have been added.
Processors
No processors have been added.


(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
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Action Maps
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Move
2D Vector
Up: <No Binding>
Down: <No Binding>
Left: <No Binding>
Right: <No Binding>
Binding Properties
Binding
Path
Composite Part
Up
Processors
No processors have been added.


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Player
Actions
Move
2D Vector
Up: <No Binding>
Down: <No Binding>
Left: <No Binding>
Right: <No Binding>
Binding Properties
Binding
Path
Compos
Listen
Usages
Gamepad
Joystick
Keyboard
Mouse
Pen
Pointer
Sensor
Touchscreen
Tracked Device
XR Controller
XR HMD
Other

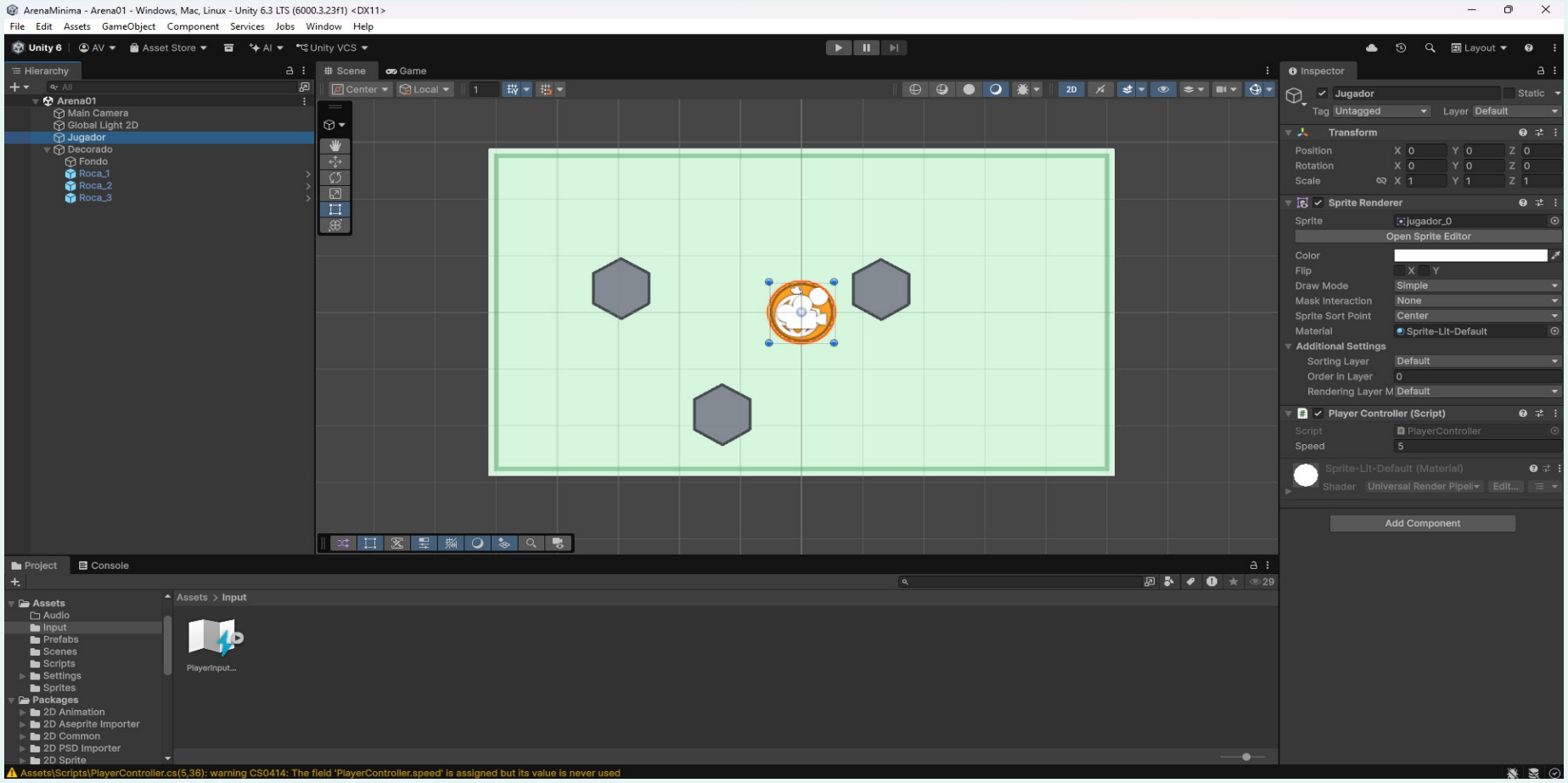

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Player
Actions
Move
2D Vector
Up: <No Binding>
Down: <No Binding>
Left: <No Binding>
Right: <No Binding>
Binding Properties
Binding
Path
Compos
Listen
Listening for Axis...
W [Keyboard]
Any Key [Keyboard]
No proce

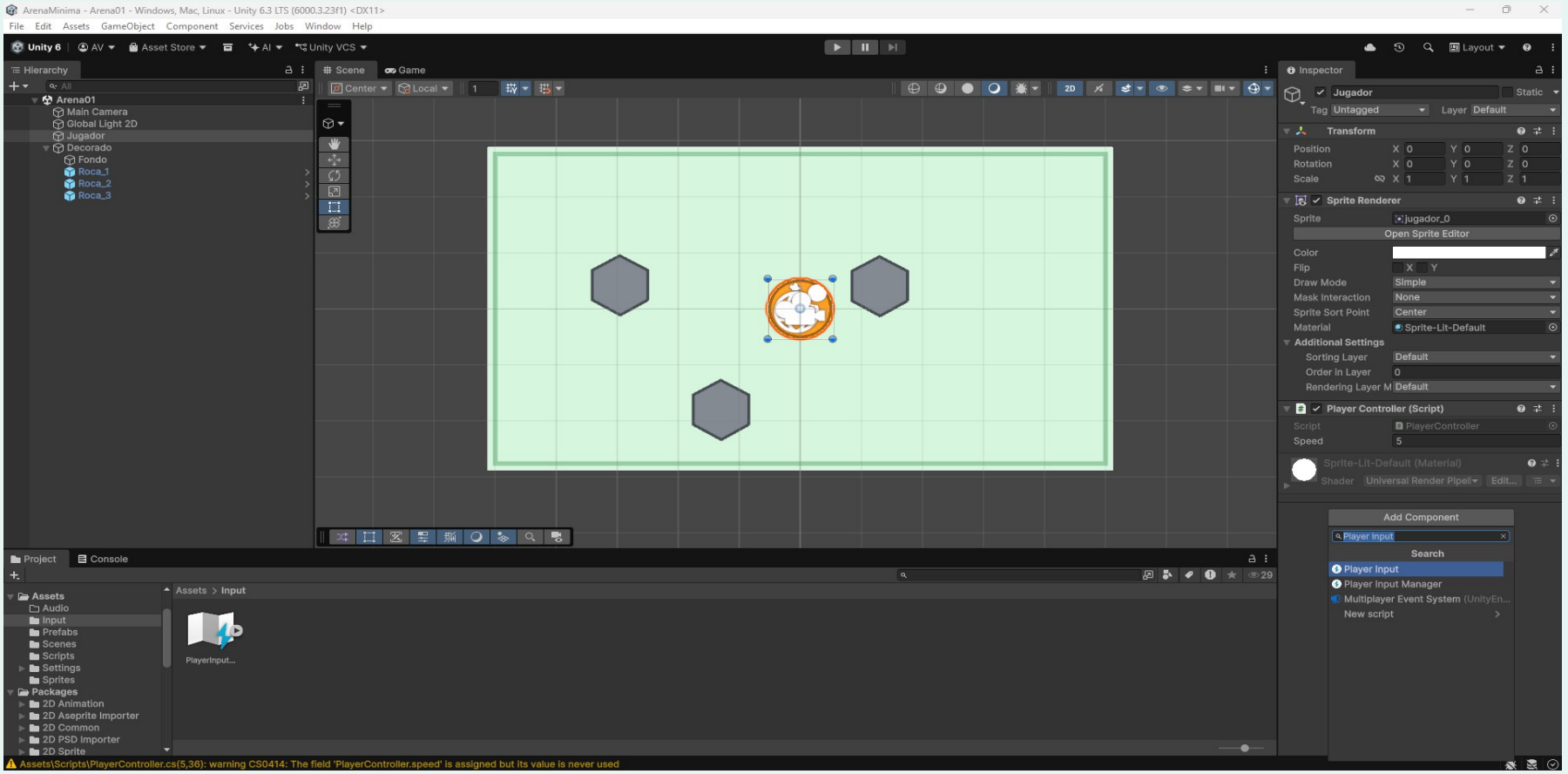

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Player
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Move
2D Vector
Up: W [Keyboard]
Down: <No Binding>
Left: <No Binding>
Right: <No Binding>
Binding Properties
Binding
Path
W [Keyboard]
T
Composite Part
Up
Processors
No processors have been added.

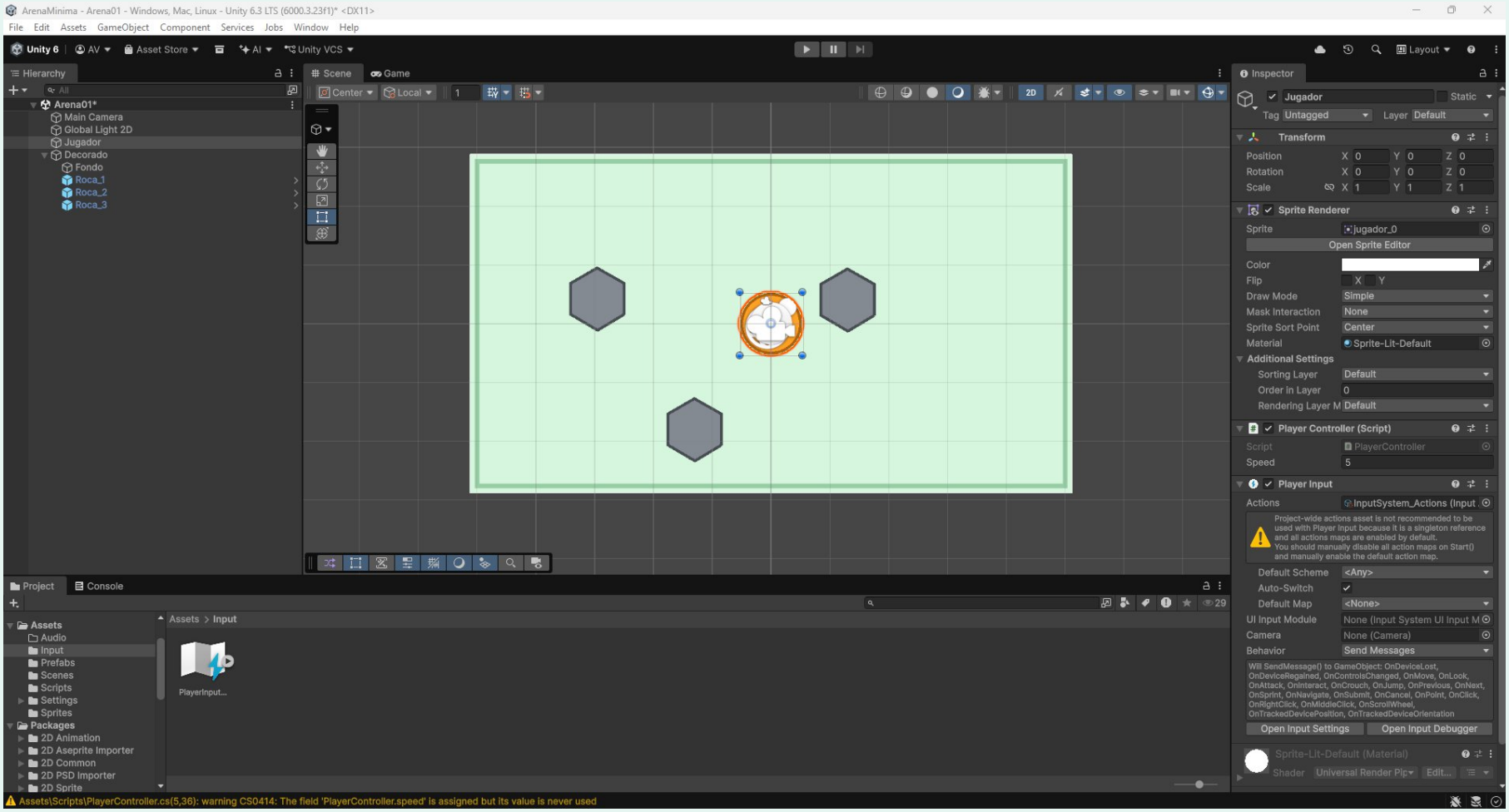

(*) PlayerInputActions (Input Actions Editor)
(*) PlayerInputActions (Input Act...
No Control Schemes
All Devices
Save Asset
Auto-Save
Action Maps
Player
Actions
Move
2D Vector
Up: W [Keyboard]
Down: S [Keyboard]
Left: A [Keyboard]
Right: D [Keyboard]
Binding Properties
Binding
Path
D [Keyboard]
T
Composite Part
Right
Processors
No processors have been added.

Configuración mínima de Move

- En Unity, en PlayerInputActions, click izquierdo en "Save Asset" y cerrar la ventana.
- En Unity, en Hierarchy, seleccionar Jugador.
- En Unity, en Inspector, click izquierdo en "Add Component" y seleccionar "Player Input".

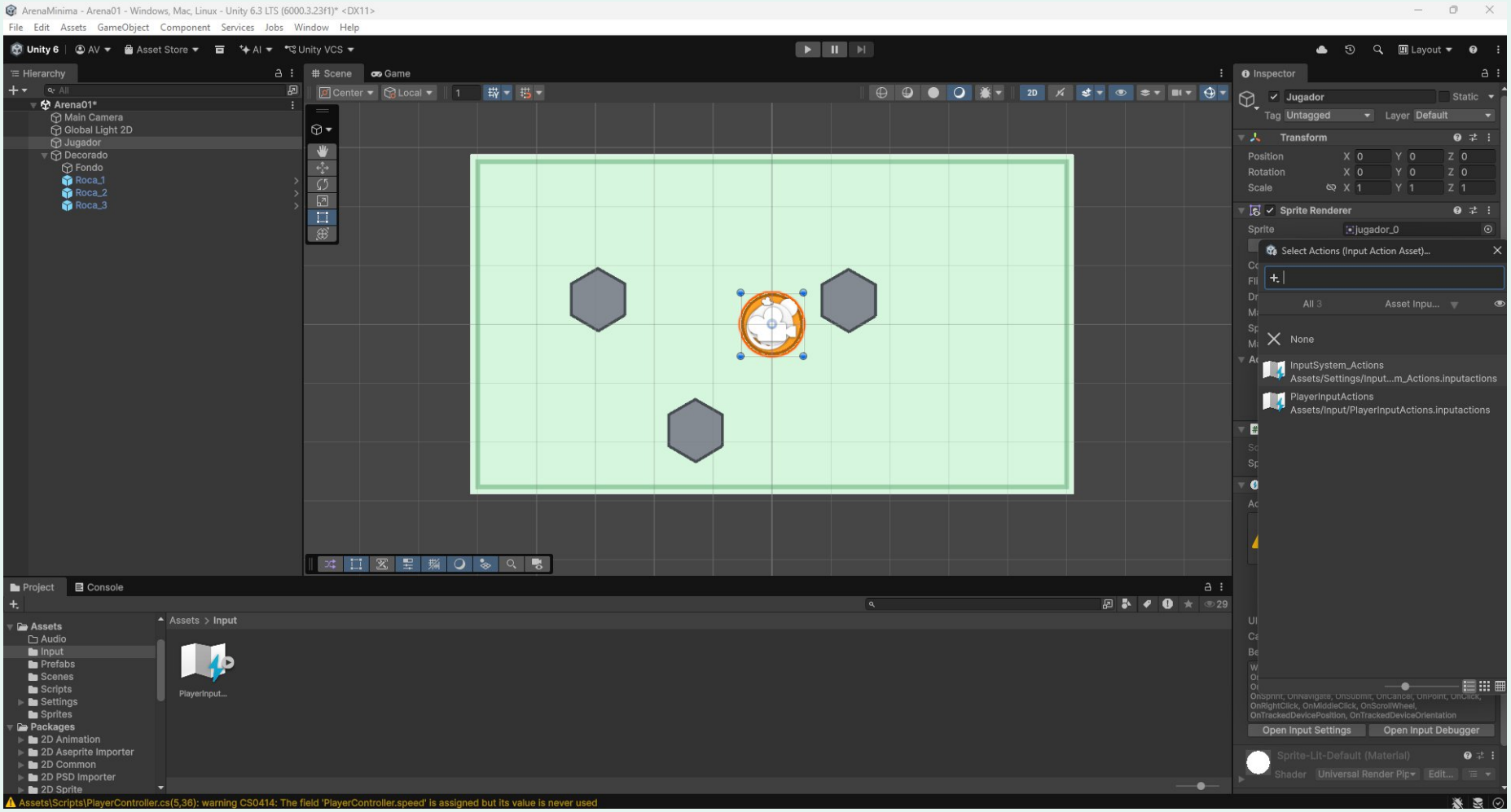

ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1) <DX11>
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Sorting Layer Default
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Player Controller (Script)
Script PlayerController
Speed 5
Sprite-Lit-Default (Material)
Shader Universal Render Pipeli
Edit...
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2D Common
2D PSD Importer
2D Sprite
Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used

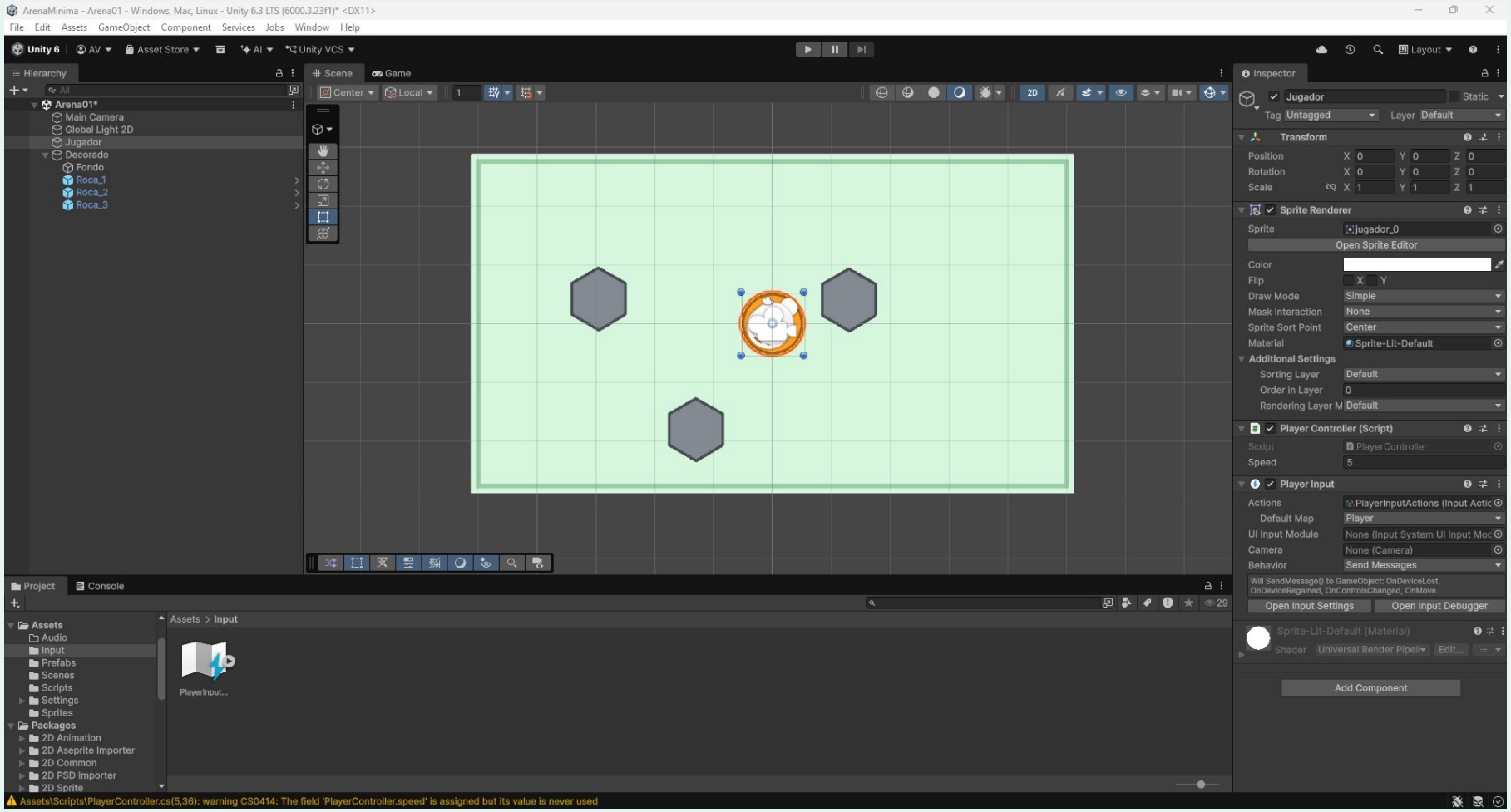

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Search
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Player Input Manager
Multiplayer Event System
New script
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ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1)* <DX11>
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Rendering Layer M Default
Player Controller (Script)
Script PlayerController
Speed 5
Player Input
Actions InputSystem_Actions (Input
Project-wide actions asset is not recommended to be used with Player Input because it is a singleton reference and all actions maps are enabled by default.
You should manually disable all action maps on Start() and manually enable the default action map.
Default Scheme <Any>
Auto-Switch
Default Map <None>
UI Input Module None (Input System UI Input M
Camera None (Camera)
Behavior Send Messages
Will SendMessage() to GameObject: OnDeviceLost, OnDeviceRegained, OnControlsChanged, OnMove, OnLook, OnAttack, OnInteract, OnCrouch, OnJump, OnPrevious, OnNext, OnSprint, OnNavigate, OnSubmit, OnCancel, OnPoint, OnClick, OnRightClick, OnMiddleClick, OnScrollWheel, OnTrackedDevicePosition, OnTrackedDeviceOrientation
Open Input Settings
Open Input Debugger
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PlayerInput...
Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used

Configuración mínima de Move

- En Unity, en Inspector, en el componente "Player Input", en Actions, asignar el asset "PlayerInputActions" que hemos creado.
- En "Default Map", comprobar que está asignado o asignar "Player".
- En "Behavior", comprobar que está asignado o asignar "Send Messages".


ArenaMinima - Arena01 - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.23f1)* <DX11>
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jugador_0
Select Actions (Input Action Asset)...
None
InputSystem_Actions
Assets/Settings/Input...m_Actions.inputactions
PlayerInputActions
Assets/Input/PlayerInputActions.inputactions
Open Input Settings
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Additional Settings
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Script PlayerController
Speed 5
Player Input
Actions PlayerInputActions (Input Actic
Default Map Player
UI Input Module None (Input System UI Input Moc
Camera None (Camera)
Behavior Send Messages
Open Input Settings
Open Input Debugger
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Assets\Scripts\PlayerController.cs(5,36): warning CS0414: The field 'PlayerController.speed' is assigned but its value is never used

Recibir Move en PlayerController

```
using UnityEngine;
using UnityEngine.InputSystem;

public class PlayerController : MonoBehaviour
{
    [SerializeField] private float speed = 3f;
    private Vector2 moveInput;

    public void OnMove(InputValue value)
    {
        moveInput = value.Get<Vector2>();
        Debug.Log("Move: " + moveInput);
    }
}
```

`Move` → `OnMove` → `Vector2`

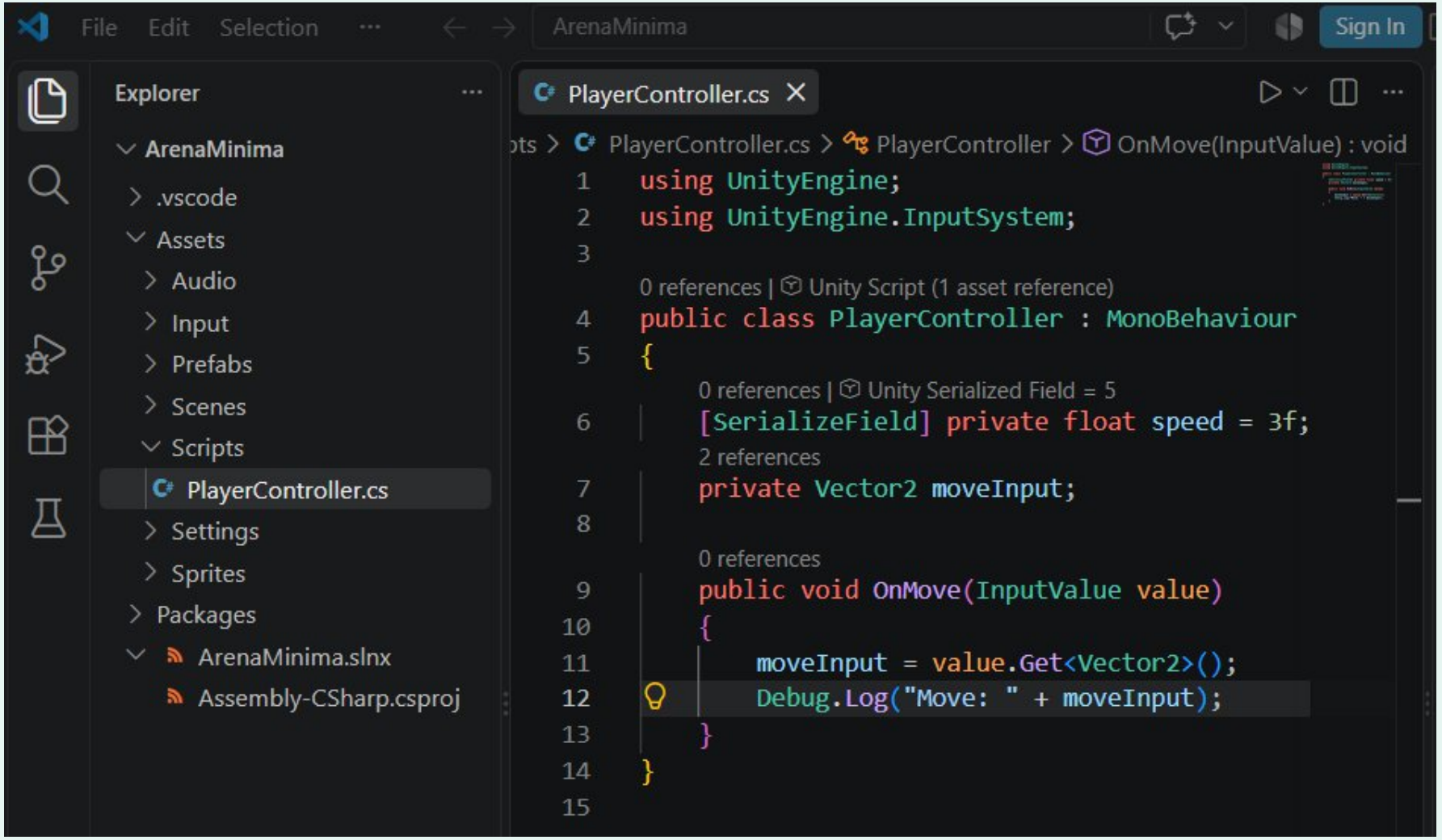
`x` → izquierda / derecha
`y` → abajo / arriba

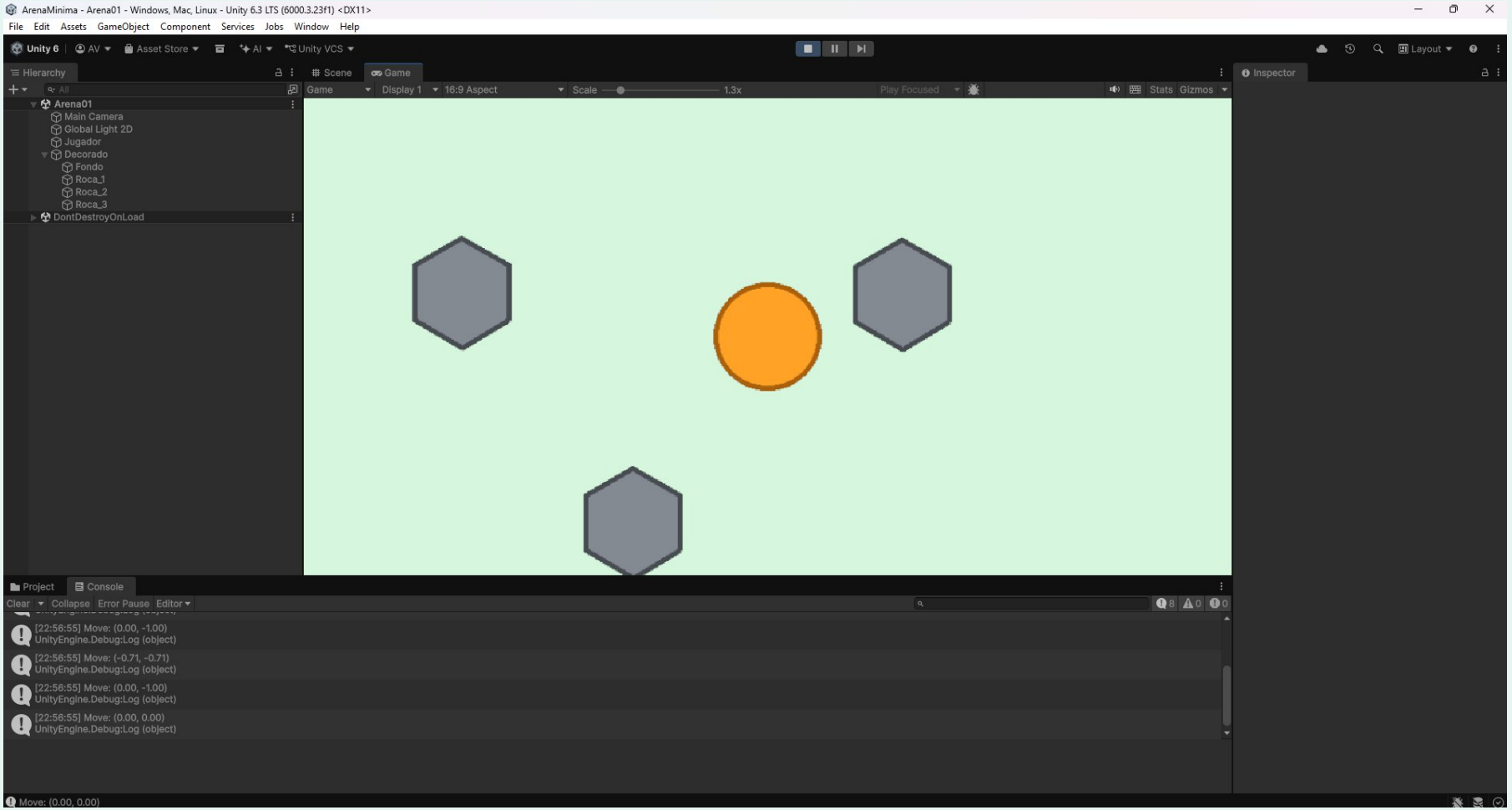



```
Console
Move: (1.00, 0.00)
Move: (0.00, 1.00)
```

Recibir Move en PlayerController

- Abrir el script PlayerController.
- Modificar el código C# conforme a la imagen.
- Guardar y cerrar VS Code.
- Volver a Unity y esperar a que compile.
- Verificar que no hay errores en Console.
- Entrar al Play Mode, probar los controles, ver el feedback vía Console y salir del Play Mode.


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Assembly-CSharp.csproj
PlayerController.cs
ots > PlayerController.cs > PlayerController > OnMove(InputValue) : void
using UnityEngine;
using UnityEngine.InputSystem;
0 references | Unity Script (1 asset reference)
public class PlayerController : MonoBehaviour
{
0 references | Unity Serialized Field = 5
[SerializeField] private float speed = 3f;
2 references
private Vector2 moveInput;
0 references
public void OnMove(InputValue value)
{
moveInput = value.Get<Vector2>();
Debug.Log("Move: " + moveInput);
}
}


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16:9 Aspect
Scale
1.3x
Play Focused
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Inspector
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Clear
Collapse
Error Pause
Editor
[22:56:55] Move: (0.00, -1.00)
UnityEngine.Debug:Log (object)
[22:56:55] Move: (-0.71, -0.71)
UnityEngine.Debug:Log (object)
[22:56:55] Move: (0.00, -1.00)
UnityEngine.Debug:Log (object)
[22:56:55] Move: (0.00, 0.00)
UnityEngine.Debug:Log (object)
Move: (0.00, 0.00)

Entrada → dirección → desplazamiento

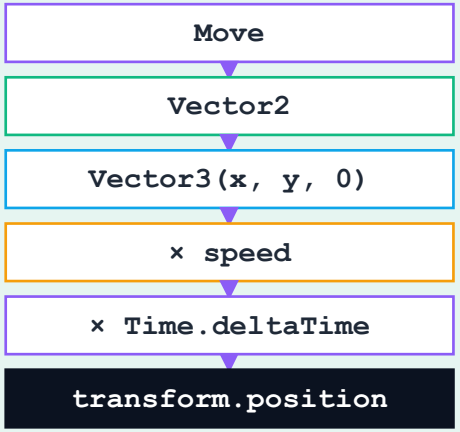
```
using UnityEngine;
using UnityEngine.InputSystem;

public class PlayerController : MonoBehaviour
{
    [SerializeField] private float speed = 3f;
    private Vector2 moveInput;

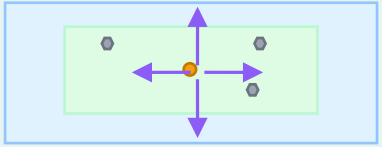
    public void OnMove(InputValue value)
    {
        moveInput = value.Get<Vector2>();
    }

    private void Update()
    {
        Vector3 direction =
            new Vector3(moveInput.x, moveInput.y, 0f);

        transform.position +=
            direction * speed * Time.deltaTime;
    }
}
```

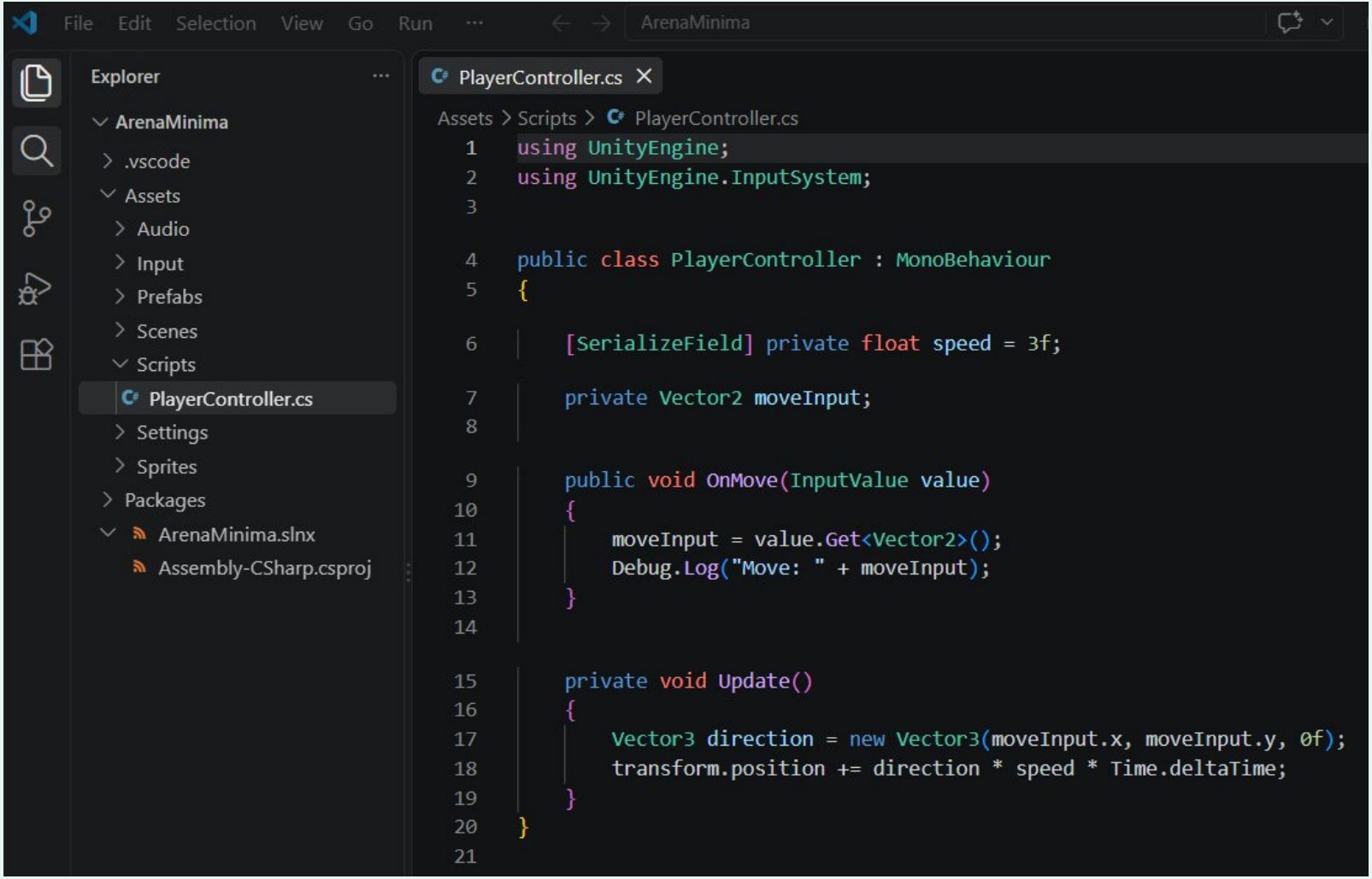


`Move` → `Vector2` → dirección → desplazamiento → Transform



Recibir Move en PlayerController

- Abrir el script PlayerController.
- Modificar el código C# conforme a la imagen.
- Guardar y cerrar VS Code.
- Volver a Unity y esperar a que compile.
- Verificar que no hay errores en Console.
- Entrar al Play Mode y comprobar que pulsando WASD se puede mover el jugador.


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PlayerController.cs
Assets > Scripts > PlayerController.cs
using UnityEngine;
using UnityEngine.InputSystem;
public class PlayerController : MonoBehaviour
{
[SerializeField] private float speed = 3f;
private Vector2 moveInput;
public void OnMove(InputValue value)
{
moveInput = value.Get<Vector2>();
Debug.Log("Move: " + moveInput);
}
private void Update()
{
Vector3 direction = new Vector3(moveInput.x, moveInput.y, 0f);
transform.position += direction * speed * Time.deltaTime;
}
}

Código, Inspector y Play Mode forman un mismo ciclo

Código

speed
OnMove
Update

Inspector

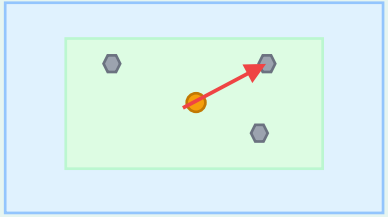
Speed
PlayerInput

Play Mode

WASD

Ajusta fuera de Play Mode

`Speed = 1` → más lento
`Speed = 6` → más rápido



Game View

`Jugador`
se desplaza

Console

sin errores
recurrentes

La Roca todavía se puede atravesar
visual ≠ físico

Depurar: no todos los fallos son iguales

Error de compilación

el código nuevo no puede compilarse correctamente

→ Console informa del problema
→ corregir antes de confiar en la prueba

Error de comportamiento

el código compila y se ejecuta

pero:
resultado obtenido ≠ resultado esperado

Micropráctica

`Speed = 0`

Play Mode:
el `Jugador` no se mueve

¿error de compilación?

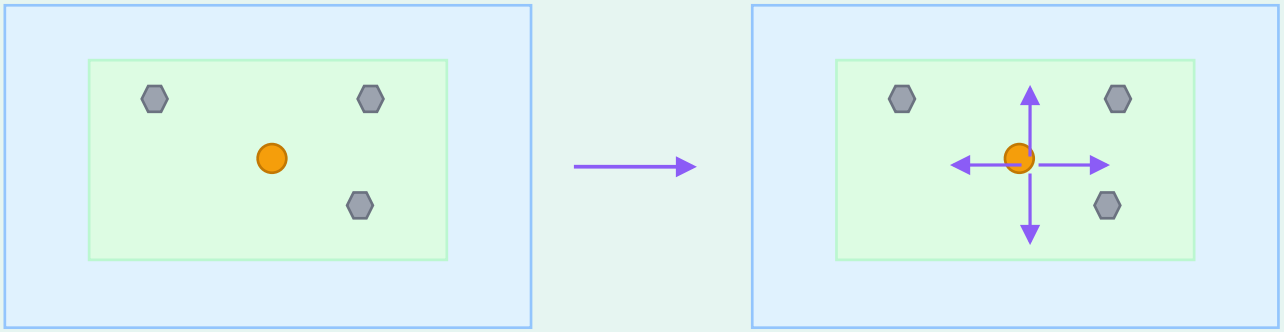
No.

Diagnóstico:
configuración incorrecta

Rutina → **modificar** → **probar** → **observar** → **diagnosticar** → **corregir** → **volver a probar**

Console no basta: también hay que observar el comportamiento.

Ya tenemos el primer prototipo interactivo



P1: `Arena01` estática

P2: `Jugador` interactivo

`Jugador`

+ PlayerController
+ Speed configurable
+ PlayerInput
+ acción `Move`

Input System
↓
Vector2
↓
Update + Time.deltaTime
↓
movimiento del `Jugador`

Todavía NO tenemos

`Rigidbody2D`
`Collider2D`
colisiones
triggers

P3

Ya podemos moverlo.

¿Cómo hacemos que el motor gestione contactos e interacciones físicas?